NONPLANAR ENTIRE MINIMIZERS OF THE FRACTIONAL ALLEN–CAHN EQUATION IN DIMENSION EIGHT

Abstract. For each $0 < s < 1/2$, we construct a nonplanar entire minimizer of the cubic fractional Allen–Cahn energy in $\mathbb{R}^8$, using the geometry and strict fractional curvature bound for an explicit cap from a companion manuscript. We transport the exact saddle to dilates of the cap and use a weighted correction near the wells to obtain strict barriers. Minimization on balls with saddle exterior data gives a limit invariant under $O(4) \times O(4)$. The limit approaches the saddle at rate $O(|P|^{-5/2})$, and its exterior zero set is a smooth normal graph of the same order over the Simons cone.

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1. Introduction

We study the fractional Allen–Cahn equation

$$(-\Delta)^s v = v - v^3 \qquad \text{in } \mathbb{R}^8, \qquad 0 < s < 1/2. \tag{1.1}$$

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The operator $(-\Delta)^s$ has Fourier symbol $|\xi|^{2s}$. An entire minimizer v has finite energy $\mathcal{E}_8(v;A)$ as defined in (2.1) for every bounded Lipschitz open set A. It minimizes this energy among measurable competitors that agree with v almost everywhere on A^c. We call v nonplanar if it cannot be written $v(P) = p(e \cdot P)$ for a unit vector e and a function p.

Write $P = (x, y) \in \mathbb{R}^4 \times \mathbb{R}^4$ and put $D = \{|y| > |x|\}$, with Simons cone $\Sigma = \partial D$. Let S be the saddle solution of (1.1) that depends only on the two block radii, is odd under interchange of the blocks, and is positive in D. Its existence and uniqueness follow from [4, Theorems 1.2 and 1.3].

Theorem 1. *For every $0 < s < 1/2$, equation (1.1) admits a smooth, sign-changing, nonplanar entire minimizer $v : \mathbb{R}^8 \to (-1, 1)$, invariant under $O(4) \times O(4)$. For some $C_s > 0$,*

$$|v(P) - S(P)| \le C_s(1 + |P|)^{-5/2} \qquad (P \in \mathbb{R}^8). \tag{1.2}$$

For each unit vector $\omega = (\omega_x, \omega_y) \notin \Sigma$, as $r \to \infty$, $v(r\omega)$ tends to 1 if $|\omega_y| > |\omega_x|$ and to -1 if $|\omega_y| < |\omega_x|$. For some $R_s > 0$ and a compact set $K_s \subset \mathbb{R}^8$, the zero set outside K_s is a smooth normal graph over $\{Z \in \Sigma : |Z| > R_s\}$, with height $O(|Z|^{-5/2})$.

For the classical Allen–Cahn equation, Liu, Wang, and Wei constructed entire minimizers whose nodal sets approach strictly area-minimizing Lawson cones in dimension eight and above, including the Simons cone in dimension eight [5, Theorem 1 in the revised author manuscript]. For $0 < s < 1/2$, Savin and Valdinoci showed that the rescaled fractional phase-transition energies have a Γ-limit proportional to the nonlocal perimeter of order $2s$ on phase indicators [8, Theorem 1.4 and Eq. (1.3)]. Dipierro, Serra, and Valdinoci proved one-dimensional symmetry for entire minimizers in dimensions $n \le 7$ and for strictly monotone solutions with directional limits ± 1 in dimension eight when s is sufficiently close to $1/2$ from below [3, Theorems 1.5–1.6 in arXiv:1611.10105v3]. Their order parameter for the operator is $2s$ in our notation. The related draft of Zhao et al. studies one-dimensional symmetry for bounded classical solutions in dimension eight that are strictly monotone in one direction, for $s_0 < s < 1/2$ with some $s_0 \in (0, 1/2)$ [9, Theorem 1.1]. The block symmetry of the minimizer constructed here gives $v(-P) = v(P)$, so its restriction to every line through the origin is even.

The geometry and strict negative curvature bound for the cap (3.1) at perimeter order $2s$ come from [11, Lemma 5.4 and Theorem 4.9]. We transport the exact saddle to a dilate of the cap using signed normal distance and meridian arclength. At the exterior cap the differential is an isometry in the meridional plane. Integration in the perpendicular directions cancels the leading change in the kernel for profiles restricted to that plane. After gluing a profile on the bounded scaled region, the cap curvature gives the leading residual on every fixed-width normal band as the dilation grows. In the pure phases a weighted correction uses the negative derivative of the cubic near the wells. Together the estimates give a global lower barrier.

The variational step follows the two-sided barrier comparison in the p-Laplace construction [10, Proposition 6.3]. We slide the lower barriers against ball minimizers with saddle exterior data. Interchanging the blocks and changing sign gives the upper bound. These bounds retain both signs under exhaustion and control the pointwise error relative to the saddle.

Section 2 establishes existence, closure, and lattice comparison for bounded-domain minimizers and fixes the normalization of the one-dimensional layer. Section 3 describes the cap and its curvature, and Section 4 gives the saddle asymptotics, well decay, and dilation estimate. Section 5 constructs the transport and estimates the exterior residual. Section 6 proves the residual estimate on bounded scaled regions, corrects the profile in the pure phases, and obtains the entire minimizer by sliding and exhaustion. Section 7 proves the ray limits, nonplanarity, and the exterior zero-set estimate.

2. Energy and comparison

Throughout the paper $0 < s < 1/2$, $\sigma = 2s$, and $L = (-\Delta)^s$ has Fourier symbol $|\xi|^{2s}$. In dimension n its kernel is $K_n(h) = c_{n,s}|h|^{-n-\sigma}$. Set

$$f(a) = a - a^3, \qquad W(a) = \frac{(1-a^2)^2}{4}, \qquad Q_A = (\mathbb{R}^n)^2 \setminus (A^c \times A^c).$$

For a bounded Lipschitz open set $A \subset \mathbb{R}^n$, define

$$\mathcal{E}_n(u; A) = \frac{1}{4}\iint_{Q_A} (u(P) - u(Q))^2 K_n(P - Q)\, dP\, dQ + \int_A W(u). \tag{2.1}$$

The pair domain consists of $A \times A$ and both orders of the cross pairs in $A \times A^c$. Exterior data are prescribed almost everywhere on A^c.

Lemma 2. *If A is a bounded Lipschitz open set, then $\int_A \int_{A^c} K_n(P - Q)\, dQ\, dP < \infty$. For exterior data in $[-1, 1]$, clipping to this interval fixes the data and does not increase the energy. There is a minimizer on A, and every minimizer takes values in $[-1, 1]$ almost everywhere. In A it has a smooth representative that solves $Lu = f(u)$ pointwise. More generally, bounded distributional solutions of this equation have uniform interior estimates of every order on compact subsets. For fixed dimension, exponent, derivative order and distance to the boundary, the estimates are uniform under a common global bound on the solutions. For any bounded measurable function locally Lipschitz in A, the absolutely convergent kernel integral represents its distributional fractional Laplacian in A. For $\varphi \in C_c^\infty(\mathbb{R}^n)$, one has $L\varphi \in L^1(\mathbb{R}^n)$.*

If u is bounded on $\mathbb{R}^n$ and $Lu = g$ in distributions on a ball, a bounded source g gives an interior C^σ bound. For $0 < \alpha < 1$ and $\alpha + \sigma \neq 1$, a C^α source gives an interior $C^{\alpha+\sigma}$ bound. On a fixed smaller concentric ball these bounds depend on the global L^∞ bound of u, the indicated norm of g on the larger ball, the exponents, the dimension, and the two radii.

Minimality on A restricts to every bounded Lipschitz open subset of A. It also passes to almost-everywhere limits on $\mathbb{R}^n$ of $[-1,1]$-valued minimizers on A, including sequences with varying exterior data.

Proof. Cross interaction and clipping. If $P \in A$ and $P+h \notin A$, the segment from P to $P+h$ meets ∂A, so $\operatorname{dist}(P,\partial A) \le |h|$. For $0 < r < 1$, the r-neighborhood of ∂A has volume at most $C_A r$. To see this, cover the compact boundary by finitely many shrunken coordinate cylinders whose closures lie in Lipschitz graph charts. For r smaller than the minimum chart margin, a nearest boundary point to any point at distance less than r lies in a shrunken cylinder, and the nearby point is still in the original chart. If the chart is $x_n = \psi(x')$, the vertical distance of the nearby point from the graph at its own horizontal coordinate is at most $(1+\operatorname{Lip}\psi)r$. This bounds the volume by the sum of the band volumes over the bounded chart bases. For the remaining $r < 1$, enlarge C_A using the finite volume of the unit neighborhood of ∂A. Using the bound $|A|$ for displacements of length at least one, we obtain

$$\begin{aligned}\int_A \int_{A^c} K_n(P-Q)\,dQ\,dP &= \int_{\mathbb{R}^n} K_n(h)\,|\{P \in A : P+h \in A^c\}|\,dh \\ &\le C_A \int_0^1 r^{-\sigma}\,dr + C|A| \int_1^\infty r^{-1-\sigma}\,dr < \infty.\end{aligned}$$

With exterior data in $[-1,1]$, the competitor equal to zero in A has finite energy by this estimate. Projection onto $[-1,1]$ fixes the datum and decreases the kinetic integrand by the 1-Lipschitz property of the projection. On the set where $|u| > 1$ it strictly decreases $W(u)$ to zero. Consequently the infimum can be computed in $[-1,1]$, and any minimizer already lies in that interval almost everywhere.

Existence and closure. We first give the limit comparison used in both claims. Suppose $u_j : \mathbb{R}^n \to [-1,1]$ converge almost everywhere to u, and each u_j minimizes on A up to an energy error $\varepsilon_j \to 0$ among competitors with its exterior datum. For any finite-energy competitor b for u on A, we may clip b to $[-1,1]$ by the preceding argument. Define $b_j = b$ on A and $b_j = u_j$ on A^c. The potential and the $A \times A$ term of b_j are those of b. On the cross pairs the integrands converge almost everywhere and are bounded by $4K_n(P-Q)$, which is integrable there. Thus each b_j has finite energy and its cross term converges to that of b by dominated convergence. Fatou's lemma and the comparison with b_j give

$$\mathcal{E}_n(u;A) \le \liminf_{j\to\infty} \mathcal{E}_n(u_j;A) \le \lim_{j\to\infty} \big(\mathcal{E}_n(b_j;A) + \varepsilon_j\big) = \mathcal{E}_n(b;A).$$

A finite-energy b exists by the constant interior construction, so this proves minimality and finiteness for the limit.

For existence, take a minimizing sequence with the fixed exterior datum and with values in $[-1,1]$. Its $L^2(A)$ norms are bounded, and the $A \times A$ energy bounds their Gagliardo seminorms in $W^{s,2}(A)$. Theorem 5.4 of [2] gives

the extension property for a Lipschitz domain, and their Theorem 7.1 with both Lebesgue exponents equal to two gives convergence of a subsequence in $L^2(A)$. Extract further for almost-everywhere convergence on A. Extend the limit by the fixed datum and apply the limit comparison. Taking $\varepsilon_j = 0$ in that comparison also proves closure for minimizers with varying data.

For restriction, let $A' \subset A$ be bounded and Lipschitz. A finite-energy competitor on A' agrees with u on $(A')^c$, so its pair integrands on $Q_A \setminus Q_{A'}$ and its potential on $A \setminus A'$ agree with those of u. These unchanged contributions are finite since u has finite energy on A. Cancelling them in the comparison on A gives the comparison on A'.

First variation. For $\varphi \in C_c^\infty(A)$, the test increment has finite squared integral against the kernel on Q_A. Indeed at least one endpoint lies in A, and the squared increment is bounded by $C\min\{|P-Q|^2, 1\}$. Integration in the displacement followed by integration over A gives finiteness. Cauchy–Schwarz gives integrability of the mixed term with the minimizer u. Since minimization holds against unrestricted competitors, we can differentiate the energy of $u + \varepsilon\varphi$ at zero. For the potential, when $|\varepsilon| \le 1$ the arguments of W stay in a fixed compact interval. The mean value theorem then bounds the potential difference quotients by $C|\varphi|$, so dominated convergence applies. Expanding the kinetic square and using $W' = -f$ gives

$$\frac{1}{2}\iint_{\mathbb{R}^n\times\mathbb{R}^n} (u(P)-u(Q))(\varphi(P)-\varphi(Q))K_n(P-Q)\,dP\,dQ = \int_A f(u)\varphi. \tag{2.2}$$

Both variables may range over $\mathbb{R}^n$, since the test difference vanishes when both lie in A^c. For $\delta > 0$, write

$$L_\delta g(P) = \int_{|P-Q|>\delta} \big(g(P) - g(Q)\big)K_n(P-Q)\,dQ.$$

For bounded measurable u, the integrable truncated kernel and $\varphi \in L^1$ justify Fubini. Symmetry gives the first equality below, and averaging the integrand with its version after interchanging P, Q gives the second.

$$\begin{aligned}\int_{\mathbb{R}^n} (L_\delta u)\varphi &= \int_{\mathbb{R}^n} uL_\delta\varphi \\ &= \frac{1}{2}\iint_{|P-Q|>\delta} (u(P)-u(Q))(\varphi(P)-\varphi(Q))K_n(P-Q)\,dP\,dQ.\end{aligned}$$

For a test function, the translation increment in L^1 is at most $|h|\|\nabla\varphi\|_{L^1}$ near zero and $2\|\varphi\|_{L^1}$ everywhere. Integrating these bounds against $K_n(h)$ gives $L\varphi \in L^1$ and, for $0 < \delta \le 1$,

$$\|L_\delta\varphi - L\varphi\|_{L^1} \le C\|\nabla\varphi\|_{L^1}\int_0^\delta r^{-\sigma}\,dr = O(\delta^{1-\sigma}).$$

For a minimizer the bilinear integrand is absolutely integrable, so the truncated pair integral tends to the full one. The pairing $\int uL_\delta\varphi$ tends to $\int uL\varphi$ by the L^1 estimate. Thus (2.2) gives $Lu = f(u)$ in distributions.

Pointwise realization. If a bounded measurable u is locally Lipschitz in A, the part deleted from $Lu(P)$ is bounded by a local Lipschitz constant times $C\int_0^\delta r^{-\sigma}\,dr$. On a compact subset of A, cover by half-radius balls on whose doubled balls u is Lipschitz. A finite subcover gives a common small radius and constant for this bound. Hence the kernel integral is absolutely convergent and $L_\delta u \to Lu$ locally uniformly in A. For fixed δ, $L_\delta u$ is a constant times u minus the convolution of u with an L^1 kernel. The convolution is continuous by continuity of translations in L^1, so $L_\delta u$ is continuous in A. The locally uniform limit is continuous. Passing to the limit in the first equality of the truncation identity identifies this pointwise integral with the distributional operator in A.

Interior regularity. The bounded-source estimate and the interior Schauder gain of order σ are given by [7, Theorem 1.1(a) and Corollary 3.5]. By evenness, the generator $-L$ has the second-increment form of that reference. Its spectral measure μ is a positive constant multiple of surface measure on S^{n-1}, as is seen by writing the kernel in polar coordinates. The mass is finite. For every unit ξ, the directional integral $\int_{S^{n-1}} |\xi\cdot\theta|^\sigma\,d\mu(\theta)$ is the same strictly positive number by rotation invariance. This gives the required uniform ellipticity, and the constant density is Hölder for every exponent in $(0,1)$. For $Lu = g$ the source for the generator $-L$ is $-g$. By translation and rescaling, the estimates on the half-radius ball have a common constant for all balls of a given radius. To estimate on B_r inside a concentric B_R, put $\ell = (R-r)/4$. The input balls $B_{2\ell}(P)$ for $P \in B_r$ lie in B_R, so the estimates on $B_\ell(P)$ have uniform constants. For the Hölder seminorm on B_r, use one of these estimates for pairs at distance less than ℓ. For pairs farther apart, use twice the uniform supremum bound for the function or derivative in the seminorm, divided by ℓ to the corresponding Hölder power. This gives the bounds for the two radii in the statement. The weak formulation of [7, Definition 2.2] is the distributional one above for bounded functions, which satisfy the growth condition in that definition.

For the nonlinear equation use $g = f(u)$. Choose $0 < b < \sigma$ so that jb and $jb+\sigma$ are nonintegers for all integers $j \ge 1$. This excludes only the countable set of values k/j and $(k-\sigma)/j$, with $k \in \mathbb{Z}$ and $j \ge 1$. The bounded-source estimate gives C^σ, and hence C^b, on a smaller ball. Suppose that u is bounded in C^{jb} on an interior ball. If $jb < 1$, the cubic gives $f(u)$ bounded in C^{jb} there. The Schauder gain gives $C^{jb+\sigma}$ on a smaller ball, from which we retain $C^{(j+1)b}$.

If $jb = k + a$ with $k \ge 1$ and $0 < a < 1$, choose a smooth cutoff χ supported in the ball of the $C^{k,a}$ estimate and equal to one on a smaller ball. On a ball compactly contained in $\{\chi = 1\}$, the cutoff equation is

$$L(\chi u)(P) = f(u(P)) + \int_{\mathbb{R}^n} (1-\chi(Q))u(Q)K_n(P-Q)\,dQ.$$

Indeed, subtract $L((1-\chi)u)$ from Lu in distributions. For a test supported in the smaller ball, the test function itself vanishes on the support of $(1-\chi)u$.

On that support its fractional Laplacian is the negative integral of the test values against the kernel. Separation of the supports and the integrable tail justify interchanging the integrals, giving the displayed positive tail term. The same separation makes the kernel derivatives in P integrable, with bounds controlled by the separation and $\|u\|_\infty$, so the term is smooth. Let β be a multi-index of order k. For a compactly supported smooth test φ, differentiating its kernel increments gives $D^\beta L\varphi = LD^\beta\varphi$. This is justified by Lipschitz bounds for each differentiated increment at short distances and boundedness in the tail. Put $w = \chi u$. Since w and $D^\beta w$ have compact support, integration by parts against $L\varphi$ is valid after multiplying it by a cutoff equal to one near their support. Thus

$$\langle LD^\beta w, \varphi\rangle = (-1)^{|\beta|}\langle w, D^\beta L\varphi\rangle = \langle D^\beta Lw, \varphi\rangle.$$

The function $D^\beta(\chi u)$ is bounded globally by the product rule. Each product uses a derivative of u controlled on the ball containing the support and a derivative of the cutoff. Its source is $D^\beta f(u)$ plus a derivative of the smooth integral. The product rule bounds $D^\beta f(u)$ in C^a, since every factor involves derivatives of u of order at most k. Applying the Schauder gain to each derivative of order k gives $C^{a+\sigma}$ regularity for those derivatives on a smaller ball. Since $\chi = 1$ there and $a + \sigma$ is not an integer, this gives $u \in C^{k+a+\sigma}$ there. Retaining order $(j + 1)b$ in either case allows the iteration to continue. For any prescribed order, choose the number of steps in advance. Use nested concentric balls with fixed positive gaps between their radii, from the initial interior radius to half that radius, reserving intermediate radii for the cutoffs. Cutoffs obtained by rescaling a fixed smooth function have derivatives bounded in terms of these gaps and the derivative orders. This yields uniform bounds on the smaller balls, hence on compact interior sets, with the stated dependence on the distance to the boundary. In the resulting smooth representative both Lu and $f(u)$ are continuous by the pointwise realization above, so the distributional equation holds everywhere in A. □

Lemma 3. *If $w : \mathbb{R}^n \to \mathbb{R}$ is bounded, measurable, and nonnegative almost everywhere, is Lipschitz near P, and satisfies $w(P) = Lw(P) = 0$, then $w = 0$ almost everywhere.*

Two minimizers on the same bounded Lipschitz open set A with the same exterior datum in $[-1, 1]$ are comparable almost everywhere.

Proof. Take a ball about P on which w is Lipschitz. Its increments at P are $O(|h|)$ near zero and bounded in the tail, so the kernel formula converges absolutely. The hypothesis $Lw(P) = 0$ uses the pointwise operator of Lemma 2. Since $w(P) = 0$,

$$0 = Lw(P) = -\int_{\mathbb{R}^n} w(Q)K_n(P - Q)\,dQ.$$

Since $w \geq 0$ and the kernel is positive, $w = 0$ almost everywhere.

For the comparison, let a, b be the two minimizers with common exterior datum $g \in [-1,1]$. The competitor equal to zero on A and g on A^c has zero $A \times A$ kinetic energy and potential energy $|A|W(0)$. Its cross term is at most $\frac{1}{2}\int_A \int_{A^c} K_n(P-Q)\,dQ\,dP$, which is finite by Lemma 2. Thus both minimizing energies are finite. The pointwise minimum and maximum of a, b have the common exterior datum. They preserve the sum of the two potential energies, since they only exchange values at each point. At pairs with the same order the kinetic sum is unchanged as well. If $a(P) > b(P)$ and $a(Q) < b(Q)$, the decrease in the sum of the two squared increments is

$$\begin{aligned}&(a(P)-a(Q))^2+(b(P)-b(Q))^2\\&\quad-(b(P)-a(Q))^2-(a(P)-b(Q))^2\\&=2(a(P)-b(P))(b(Q)-a(Q))>0.\end{aligned}\tag{2.3}$$

Interchanging P and Q covers the other opposite order. The kinetic sum never increases, so both competitors have finite energy. If $\{a > b\}$ and $\{a < b\}$ both had positive measure, they would lie in A up to null sets. Their product would contribute a strictly positive decrement to the kinetic sum, by (2.3) and positivity of K_n. The sum of the two minimizing inequalities forbids this. Thus one of the strict-order sets has measure zero. □

Kernel marginals and the one-dimensional layer. For a coordinate split $h = (\xi, \zeta) \in \mathbb{R}^k \times \mathbb{R}^{n-k}$, $1 \le k < n$, the normalized kernel satisfies

$$\int_{\mathbb{R}^{n-k}} K_n(\xi,\zeta)\,d\zeta = K_k(\xi) \qquad (\xi \neq 0).\tag{2.4}$$

In fact, scaling ζ by $|\xi|$ gives a positive constant times $|\xi|^{-k-\sigma}$. To identify that constant, evaluate the Fourier symbol at the unit vector in the first retained coordinate. In dimension j the normalization reads

$$\int_{\mathbb{R}^j} (1-\cos h_1) K_j(h)\,dh = 1.$$

The cosine appears by evenness of the kernel, and the integral converges by the quadratic increment at zero and the integrable tail. Tonelli's theorem applied to the same identity in dimension n fixes the constant in the marginal to be $c_{k,s}$. In particular, L applied to a bounded smooth function of one coordinate reduces to the one-dimensional L, by (2.4) and absolute convergence for $\sigma < 1$.

To apply [1, Theorem 2.4], take a smooth compactly supported cutoff of f, denoted by $\widetilde{f}$, equal to f on a neighborhood of $[-1,1]$. Its derivative is globally Hölder with every exponent in $(1-\sigma, 1)$. The potential $G(u) = -\int_0^u \widetilde{f}(z)\,dz$ differs from W by a constant on $[-1,1]$, since $W' = -f$. Thus $G > G(-1) = G(1)$ on $(-1,1)$ and $G'(\pm 1) = 0$. Also $\widetilde{f}'(\pm 1) = f'(\pm 1) = -2$, so the theorem gives a strictly increasing layer with limits ± 1, unique up to translation. Every strictly increasing profile with these limits takes values in $(-1,1)$, where $\widetilde{f} = f$. This identifies the layer class for the cutoff with

that for the cubic. The limits and strict increase give a unique zero, and we translate the layer so that $q(0) = 0$.

By [1, Lemma 2.1] the bounded layer is locally $C^{2,\alpha}$ for some $\alpha > 0$, since the cutoff satisfies the preceding Hölder condition. The pointwise equation therefore holds in distributions by Lemma 2. Its interior estimates on translated unit intervals give smoothness and uniform bounds on the derivatives of q.

3. The cap and its orientation

For $P = (x, y) \in \mathbb{R}^4 \times \mathbb{R}^4$, write $a = |x|$, $b = |y|$, and set

$$D = \{b > a\}, \qquad \Sigma = \partial D, \qquad \rho = \frac{a+b}{\sqrt{2}}, \qquad t = \frac{b-a}{\sqrt{2}}.$$

By continuity, ∂D lies in $\{b = a\}$. Conversely, at any point with $b = a > 0$, slightly enlarging either block gives points on the two sides. At the origin use small vectors in the two factor spaces separately. Thus $\Sigma = \{b = a\}$. The inverse formulas $a = (\rho - t)/\sqrt{2}$ and $b = (\rho + t)/\sqrt{2}$ give the admissible region $\rho \geq |t|$. Since $|P|^2 = \rho^2 + t^2$, we have $|P|/\sqrt{2} \leq \rho \leq |P|$. The map $P \mapsto (a, b)$ is 1-Lipschitz by the reverse triangle inequality in each block. Its two unit linear projections ρ and t are therefore 1-Lipschitz as well. For any fixed block radii $c, e \geq 0$, the reverse triangle inequality bounds the squared distance from P to a point with those radii below by $(a - c)^2 + (b - e)^2$. Equality is attained by matching the direction of each nonzero block of P, with arbitrary direction for a zero block. On Σ the two radii coincide, so we minimize over their common value $u \geq 0$. The resulting squared distance at fixed u is

$$(a-u)^2 + (b-u)^2 = 2\left(u - \frac{a+b}{2}\right)^2 + \frac{(b-a)^2}{2}.$$

Thus t is the signed Euclidean distance to Σ, positive on D.

For a measurable set G, we use the unnormalized exterior-minus-interior convention

$$H_\sigma[G](p) = \lim_{\varepsilon \downarrow 0} \int_{|Q-p|>\varepsilon} \frac{\mathbf{1}_{G^c}(Q) - \mathbf{1}_G(Q)}{|p - Q|^{8+\sigma}}\, dQ$$

when the limit exists. Every fixed truncation is absolutely integrable, since the integral of the kernel is $|S^7| \int_\varepsilon^\infty r^{-1-\sigma}\, dr$. At a point $p = (x_0, y_0) \in \Sigma \setminus \{0\}$, choose an orthogonal map R of $\mathbb{R}^4$ with $Rx_0 = y_0$. The transformation $(x, y) \mapsto (R^{-1}y, Rx)$ is an isometry that fixes p and interchanges the two open chambers. Under this change of variables the kernel and every centered truncation are unchanged, while $\mathbf{1}_{D^c} - \mathbf{1}_D$ changes sign away from Σ. The cone has measure zero by Fubini, since for each fixed x the section in the y-block is the sphere of radius $|x|$. The change of variables therefore gives zero at every truncation, and $H_\sigma[D](p) = 0$.

The cap is

$$h(r) = (1 + 2r^2/5)^{-3/8}, \qquad E = \{|y|^2 - |x|^2 > h(2|x||y|)\}. \tag{3.1}$$

The geometry lemma in [11, Lemma 5.4] gives the smooth boundary $b = h_*(a)$. Here $h_*(0) = 1$, $h_*(a) > a$, and $0 < h_*'(a) < 1$ for $a > 0$. The function h_* is smooth and even at zero. For each fixed a, the defining function $b^2 - a^2 - h(2ab)$ is strictly increasing in $b \geq 0$, since b^2 increases strictly and $h(2ab)$ is nonincreasing. Continuity and the boundary description give a zero at $b = h_*(a)$, so $E = \{b > h_*(a)\}$. At $x = 0$, $|y| = 1$ the defining function $b^2 - a^2 - h(2ab)$ has Cartesian gradient $(0, 2y)$, since $h(2ab) = (1 + 8a^2b^2/5)^{-3/8}$. As the defining function is positive in E, the inward unit normal there is $(0, y)$.

Along the boundary, $\rho = (a+h_*(a))/\sqrt{2}$ starts at $\rho_{\mathrm{ax}} = 1/\sqrt{2}$ and tends to infinity because $h_*(a) > a$. For $a > 0$ its derivative is $(1+h_*'(a))/\sqrt{2} > 0$, so the inverse is smooth by the inverse function theorem. Thus the boundary is $t = \eta(\rho)$ in these coordinates, with $\eta(\rho_{\mathrm{ax}}) = \rho_{\mathrm{ax}}$ and $\eta(\rho) > 0$. Since $t = (h_*(a) - a)/\sqrt{2}$ on the boundary,

$$\eta'(\rho) = \frac{h_*'(a) - 1}{h_*'(a) + 1} < 0 \qquad (\rho > \rho_{\mathrm{ax}}).$$

At fixed ρ, increasing t increases b and decreases a, so $b - h_*(a)$ increases strictly. For $\rho > \rho_{\mathrm{ax}}$ it vanishes at $t = \eta(\rho)$, so E is given by $t > \eta(\rho)$ in that admissible region. For $\rho \leq \rho_{\mathrm{ax}}$, $b \leq \sqrt{2}\rho \leq 1 \leq h_*(a)$, so there are no points of E there. In the plane obtained by fixing the two block directions, the inward unit normal to the graph at $\rho > \rho_{\mathrm{ax}}$ is

$$\nu = \frac{(-\eta'(\rho), 1)}{\sqrt{1 + \eta'(\rho)^2}}.$$

This is the direction of increase of the defining function $t - \eta(\rho)$. Invariance under block rotations makes the normal orthogonal to the angular directions. Also, $\operatorname{dist}(0, \overline{E}) = 1$, since $b \geq h_*(a) \geq 1$ on $\overline{E}$ and $(0, y) \in \partial E$ when $|y| = 1$. On $\overline{E} \cap \overline{B}_R$ for $R > 0$, continuity extends the defining inequality to $2\rho t \geq h(\rho^2 - t^2)$. Since $0 \leq \rho^2 - t^2 = 2ab \leq |P|^2 \leq R^2$ and h is decreasing, the right-hand side is at least $h(R^2) > 0$. Using $\rho \leq R$ yields $t \geq h(R^2)/(2R)$ there, giving positive separation from the cone on each bounded portion of the cap.

The coordinates (ρ, t) are orthonormal in each fixed block-direction plane. The cone is $t = 0$ in this plane, and its outward normal points toward decreasing t. Thus the height in the geometry lemma is $-\eta$. With $\kappa = 5/2$ and $B = (5/2)^{3/8}/2$, its expansion gives, as $\rho \to \infty$,

$$\eta(\rho) = B\rho^{-\kappa} - \frac{15B}{16}\rho^{-\kappa-4} + O_5(\rho^{-\kappa-7}). \tag{3.2}$$

The notation $O_5(\rho^{-\kappa-7})$ means that the j-th derivative of the remainder is $O(\rho^{-\kappa-7-j})$ for $0 \leq j \leq 5$. In particular, at sufficiently large ρ we have $\eta(\rho) \asymp \rho^{-\kappa}$ and $|\eta^{(j)}(\rho)| \leq C_j\rho^{-\kappa-j}$ for $1 \leq j \leq 3$. The source uses the same curvature convention and denotes the perimeter order by s. Taking that order to be our $\sigma = 2s \in (0, 1)$, its geometry lemma and strict-sign

theorem [11, Lemma 5.4 and Theorem 4.9] give

$$H_\sigma[E](p) \le -c_\sigma(1+\rho(p))^{-1-\sigma-\kappa}, \qquad p \in \partial E, \tag{3.3}$$

for a constant $c_\sigma > 0$, including at the axis. Because $\rho(p) \le |p|$, the bound also holds with $|p|$ in place of $\rho(p)$.

For $E_\lambda = \lambda E$, both ρ and t scale by λ, so $\eta_\lambda(\rho) = \lambda\eta(\rho/\lambda)$. For any measurable set G, whenever $H_\sigma[G](p)$ exists, one has

$$H_\sigma[\lambda G](\lambda p) = \lambda^{-\sigma} H_\sigma[G](p) \qquad (\lambda > 0). \tag{3.4}$$

Indeed $Q = \lambda Y$ multiplies the truncated integral by $\lambda^{-\sigma}$ and changes the deleted radius from ε to ε/λ.

We will also use a local estimate when G occupies one side of a smooth hypersurface near $p \in \partial G$, with inward unit normal ν. In a tubular neighborhood the signed distance, taken positive on G, is smooth. Its gradient at p is ν, since the distance vanishes along the interface and increases at unit speed on the inward normal. Taylor's formula gives $\nu \cdot h + O(|h|^2)$ at $p + h$. The remainder is bounded by $M|h|^2$ on a ball about p, with M determined by the signed-distance Hessian bound. Thus $2\mathbf{1}_G(p+h) - 1$ agrees with $\operatorname{sgn}(\nu \cdot h)$ outside the slab $|\nu \cdot h| \le M|h|^2$, and their difference is at most 2 in absolute value.

For a uniform point on the unit sphere S^{n-1}, $n \ge 3$, the coordinate in any unit direction has density proportional to $(1-u^2)^{(n-3)/2}$ on $(-1,1)$. Indeed the parametrization $(u, \sqrt{1-u^2}\,\omega)$, $\omega \in S^{n-2}$, scales the $n-2$ angular directions by $\sqrt{1-u^2}$. The u-derivative is orthogonal to those directions and has length $1/\sqrt{1-u^2}$. The surface element is therefore $(1-u^2)^{(n-3)/2}\,du\,dS(\omega)$. It gives a bounded density for u and a uniform conditional angular law on S^{n-2} at fixed u. For $h = r\omega$ the slab requires $|\nu \cdot \omega| \le Mr$, so its angular measure is bounded by CMr. In dimension eight, for ε within the ball of the Taylor estimate, polar integration gives

$$\begin{aligned} &\int_{|h|<\varepsilon} \frac{\bigl|2\mathbf{1}_G(p+h) - 1 - \operatorname{sgn}(\nu\cdot h)\bigr|}{|h|^{8+\sigma}}\,dh \\ &\qquad \le 2\int_{|h|<\varepsilon} \frac{\mathbf{1}_{\{|\nu\cdot h| \le M|h|^2\}}}{|h|^{8+\sigma}}\,dh \le CM\int_0^\varepsilon r^{-\sigma}\,dr = \frac{CM}{1-\sigma}\varepsilon^{1-\sigma}. \end{aligned} \tag{3.5}$$

For $0 < \delta < \varepsilon$, the difference of the two curvature truncations is the integral of $1 - 2\mathbf{1}_G(p+h)$ against the kernel on $\delta < |h| \le \varepsilon$. The tangent half-space sign integrates to zero there by oddness, so (3.5) bounds the absolute value of the difference. Since $1 - \sigma > 0$, the truncations are Cauchy as the deleted radius tends to zero. Letting $\delta \downarrow 0$ gives the same bound for the error between the principal value and the truncation at ε. Both bounds are uniform under a common Hessian bound and a common radius for the Taylor estimate. Figure 1 illustrates the choice of inward normal.

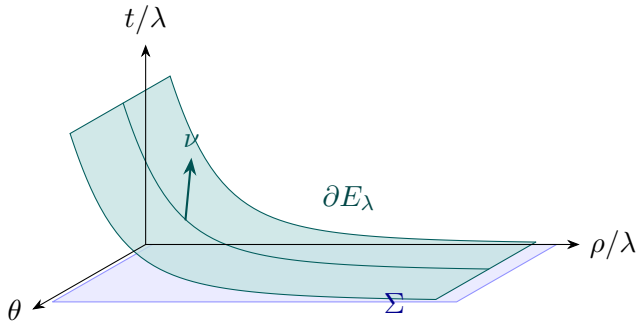


FIGURE 1. Projection of the cone $t = 0$ and the cap boundary $t = \eta_\lambda(\rho)$ in the scaled coordinates $(\rho/\lambda, t/\lambda)$, with θ a local angle in the y-block. The arrow represents the inward unit normal ν in the (ρ, t)-plane, pointing toward the side $t > \eta_\lambda(\rho)$.

4. Estimates for the eight-dimensional saddle

Use the coordinates a, b, ρ, t of Section 3, and write $d_\Sigma = t$ and $T(x, y) = (y, x)$.

Lemma 4 (Differentiation and decay near the wells)**.** *For a bounded smooth solution of $Lv = f(v)$ in $\mathbb{R}^n$, each coordinate derivative satisfies $L(\partial_j v) = f'(v)\partial_j v$. If $n = 8$ and $v(X) \to \pm 1$ uniformly as $d_\Sigma(X) \to \pm\infty$, respectively, then*

$$\big|\, 1 - |v(X)| \,\big| + |\nabla v(X)| \le C(1 + |d_\Sigma(X)|)^{-\sigma} \qquad (X \in \mathbb{R}^8).$$

Proof. By Lemma 2, the pointwise equation holds in distributions, and the interior estimates on translated unit balls bound ∇v and $D^2 v$ uniformly. For $\tau \neq 0$ set $v_\tau(P) = (v(P + \tau e_j) - v(P))/\tau$, where e_j is the j-th coordinate vector. This quotient is the average of $\partial_j v$ along the segment from P to $P + \tau e_j$. In particular,

$$|v_\tau(P + k) - v_\tau(P)| \le \min\{\|D^2 v\|_\infty |k|,\, 2\|\nabla v\|_\infty\}.$$

The bound is integrable against $K_n(k)$ because $0 < \sigma < 1$. Translation invariance gives Lv_τ as the same difference quotient of $f(v)$. Letting $\tau \to 0$, dominated convergence on the kernel side and the chain rule on the reaction side give $L(\partial_j v) = f'(v)\partial_j v$.

Fix X with $D_0 = d_\Sigma(X) > 0$ and let $R = D_0/2$. By the Lipschitz bound from Section 3, $d_\Sigma \ge D_0/2$ on $B_R(X)$. The positive end condition therefore gives $v \ge 3/4$ on the whole ball for large D_0. On that ball,

$$(L + v(1 + v))(1 - v) = 0, \qquad (L + 3v^2 - 1)\partial_j v = 0.$$

The first identity follows from $f(v) = v(1 + v)(1 - v)$, and the second from $f'(v) = 1 - 3v^2$. Both coefficients are at least $c_* = 11/16 > 0$. The equations

are linear in $1 - v$ and $\partial_j v$, so the same barrier works for these functions and their negatives.

Choose a smooth function $0 \le \zeta \le 1$ which vanishes near zero and is one outside B_1. The kernel integral of $L\zeta$ is bounded by using the Lipschitz increment at distances below one and the bounded increment in the tail. Let M exceed the global bounds of the functions being compared. Set

$$B(P) = AR^{-\sigma} + M\zeta((P - X)/R), \qquad A \ge M\|L\zeta\|_\infty / c_*.$$

The barrier dominates each compared function outside $B_R(X)$. By scaling, $LB(P) = MR^{-\sigma}(L\zeta)((P - X)/R)$. For either coefficient $c(P) \ge c_*$, this gives

$$(L + c(P))B(P) \ge R^{-\sigma}\big(c_* A - M\|L\zeta\|_\infty\big) \ge 0.$$

If a compared function u exceeded B, then $u - B$ would attain a positive global maximum inside the ball. At that point $L(u - B) \ge 0$ by positivity of the kernel and $c(P)(u - B) > 0$, contrary to the equation and the barrier inequality. At the center X we obtain $|1 - v(X)| + |\nabla v(X)| \le CD_0^{-\sigma}$. If $d_\Sigma(X) < 0$, use $-v$ and $-d_\Sigma$ in the same argument. The oddness of f gives the equation for $-v$, and the negative end condition gives $-v \ge 3/4$ on the corresponding ball for large $|d_\Sigma(X)|$. On either end v has the prescribed sign, so the estimates give the claimed absolute well error. On the remaining region of bounded $|d_\Sigma|$, the global bounds on v and ∇v suffice. □

Apply [4, Theorem 1.2(i)–(ii)] with two blocks of dimension four and exponent $\gamma = s$. Their trace operator is the fractional Laplacian of Section 2. The hypotheses hold for the smooth odd cubic, which satisfies $f(0) = f(1) = 0$ and $f'' < 0$ on $(0, 1)$. Order the two blocks so that the radial variables (s, t) of that reference correspond to (b, a) here. The resulting saddle S is positive in D, invariant under the two block rotations, and odd under T, by their Definition 1.1. It takes values in $(-1, 1)$. On Σ the two radii are equal, so swapping them leaves the value of S unchanged. Oddness under the swap then gives $S = 0$ on all of Σ. Lemma 2 gives smoothness and uniform derivative bounds on translated unit balls. The Dirichlet-to-Neumann identity preceding (1.3) of [4] gives the one-dimensional equation $Lu_0(\cdot, 0) = f(u_0(\cdot, 0))$ for the trace of their extension layer. Its strictly positive derivative, end values ± 1, and centering in (1.5) identify it with q by the uniqueness in Section 2. Theorem 1.2(ii) of [4] is stated for the fractional equation (1.1) on $\mathbb{R}^{2m}$. With their convention after (1.3), u and U in its $L^\infty(\mathbb{R}^{2m} \setminus B_R)$ estimate denote the traces, and $\nabla_x(u - U)$ is their spatial gradient difference. Here $u = S$ and the trace of the profile (1.6) is $q \circ d_\Sigma$. Thus the estimate gives

$$S(X) - q(d_\Sigma(X)) \longrightarrow 0 \quad \text{uniformly as } |X| \to \infty,$$

and the same uniform convergence for the spatial gradient off the factor axes, where $q \circ d_\Sigma$ is smooth. The essential supremum in the cited estimate gives a pointwise bound there by continuity. Since $|d_\Sigma(X)| \le |X|$, the convergence of the values and the limits of q give the uniform end conditions

in Lemma 4. We will also use $\inf_{\{d_\Sigma \geq d_0\}} S > 0$ for every $d_0 > 0$. Indeed, the value asymptotics give $S \geq q(d_0)/2$ outside a sufficiently large ball in this region. Its intersection with the closed ball is a compact subset of D, where $S > 0$.

Write $S = S(\rho, d)$ off the factor axes, with $d = d_\Sigma$. In D off the axes, ρ is the variable y in [4, Proposition 1.8(i)–(ii)], and a is their smaller radius. Their trace inequalities give $S_\rho \geq 0$ and $S_a < 0$, where S_a is the derivative in a at fixed b. The coordinate identities $\sqrt{2}S_a = S_\rho - S_d$ and $\sqrt{2}S_b = S_\rho + S_d$ give $S_d > 0$ and $S_b > 0$ there. The identity $S(a, b) = -S(b, a)$ gives $S_a(a, b) = -S_b(b, a)$ and $S_b(a, b) = -S_a(b, a)$, so the same block derivative signs hold in the opposite chamber off the axes. These signs give weak block monotonicity throughout the quadrant by continuity, including on the cone and the axes. In both open chambers off the axes, $S_d = (S_b - S_a)/\sqrt{2} > 0$. At fixed $\rho > 0$, integrating on the two sides of $d = 0$ and using continuity at $d = 0$ gives strict increase on $(-\rho, \rho)$.

Lemma 5 (Dilation derivative of the saddle)**.** *The function $P(X) = X \cdot \nabla S(X)$ and its first derivatives are bounded in $\mathbb{R}^8$. In particular, off the factor axes,*

$$|S_\rho(\rho, d)| \leq \frac{C}{\rho} \qquad (|d| < \rho),$$

and for each $M > 0$,

$$|S_{\rho d}(\rho, d)| \leq \frac{C_M}{\rho} \qquad (|d| \leq M,\ \rho \geq 2M + 1).$$

Proof. We will compare $S(\lambda X)$ to $S(X)$ for λ just above one. The comparison uses a positive function for the linearized equation on the odd class. We first derive the angular kernel bounds that give its strict sign, including at small d.

Angular averaging. Let $0 \leq a < b$ denote the block radii of $X \in D$, and $0 \leq c < e$ those of an integration point in D. Use probability measure $d\bar{\omega}$ on each copy of S^3 and radial measure $d\mu = |S^3|^2 c^3 e^3 \, dc \, de$. Let K_{d} be the angular average of $c_{8,s}|X - Y|^{-8-\sigma}$, and let K_{c} be the same average with the two integration radii interchanged. Set $J = K_{\mathrm{d}} - K_{\mathrm{c}}$. For a bounded block-invariant u odd under T and locally Lipschitz in D, the integral at $X \in D$ converges absolutely. The local Lipschitz bound controls the singularity, and X is at positive distance from the opposite chamber. We can thus split the integral into the two chambers, discarding the measure-zero cone, and change variables by T in the opposite chamber. In radial variables this gives

$$\begin{aligned} L_{\mathrm{o}} u(X) &= \int_{c<e} (u(X) - u(c, e)) J(a, b; c, e) \, d\mu + A(X) u(X), \\ A(X) &= 2 \int_{c<e} K_{\mathrm{c}}(a, b; c, e) \, d\mu. \end{aligned} \tag{4.1}$$

To see the multiplication term, the integral over the opposite chamber becomes $(u(X) + u(c, e))K_{\mathrm{c}}$ after the swap. Adding the direct term $(u(X) -$

$u(c,e))K_{\mathrm{d}}$ gives the displayed formula. In Cartesian coordinates, $A(X) = 2\int_{D^c} K_8(X-Y)\,dY$. This is finite by the separation from D^c and positive because the kernel is positive on a set of positive measure in the opposite chamber.

Set $B(z) = \int_{S^3} e^{z\omega_1}\,d\bar{\omega}$ and $p = (8+\sigma)/2$. By

$$w^{-p} = \frac{1}{\Gamma(p)}\int_0^\infty e^{-\tau w}\tau^{p-1}\,d\tau,$$

we can average the Gaussian first by Tonelli's theorem. Rotate the two blocks so that $X = (ae_1, be_1)$ and write the direct integration point as $Y = (c\omega, e\omega')$. Then $|X-Y|^2 = a^2+b^2+c^2+e^2-2ac\omega_1-2be\omega_1'$, so the angular integrals factor. The factors for the direct and cross kernels at $\tau > 0$ are

$$M_{\mathrm{d}} = e^{-\tau(a^2+b^2+c^2+e^2)}B(2\tau ac)B(2\tau be),$$
$$M_{\mathrm{c}} = e^{-\tau(a^2+b^2+c^2+e^2)}B(2\tau ae)B(2\tau bc).$$

Their logarithmic ratio is

$$\log\frac{M_{\mathrm{d}}}{M_{\mathrm{c}}} = \int_a^b\int_c^e 2\tau\big[(\log B)'(2\tau uv) + 2\tau uv\,(\log B)''(2\tau uv)\big]\,dv\,du. \tag{4.2}$$

Indeed the logarithm of the ratio is the rectangular difference of the function $(u,v)\mapsto \log B(2\tau uv)$, so the fundamental theorem of calculus integrates its mixed derivative. Here $(\log B)'(0) = 0$ by reflection on the sphere. At any z, $(\log B)''(z) = B''(z)/B(z) - (B'(z)/B(z))^2$ is the variance of ω_1 under the probability measure $e^{z\omega_1}d\bar{\omega}/B(z)$. The variance is positive because this measure has positive density and ω_1 is nonconstant. In particular $(\log B)'(z) \geq 0$ for $z \geq 0$, so the integrand in (4.2) is nonnegative. For $(a,b) \neq (c,e)$, the block-distance bound from Section 3 gives a positive lower bound for the distance from X to every direct integration point with radii c, e. The cross integration points also stay a positive distance away. Both kernel averages are therefore finite, and the Gaussian representation gives

$$J(a,b;c,e) = \frac{c_{8,s}}{\Gamma(p)}\int_0^\infty (M_{\mathrm{d}} - M_{\mathrm{c}})\tau^{p-1}\,d\tau \geq 0.$$

The omitted pair $(c,e) = (a,b)$ has $d\mu$-measure zero.

We need two lower bounds, both obtained by restricting the Gaussian integral to $1 \leq \tau \leq 2$. If a, b are bounded and (c,e) runs in a compact rectangle contained in $0 < c < e$, the variance has a positive lower bound on the bounded range of $2\tau uv$. Integrating $(\log B)''$ from zero gives $(\log B)'(2\tau uv) \geq c_1 uv$. The u, v integrals of uv give $(b^2-a^2)(e^2-c^2)/4$, with e^2-c^2 bounded below on the rectangle. Thus the logarithmic ratio is at least $c_2(b^2-a^2)$. Since M_{c} has a positive minimum on these compact parameter sets, writing $\ell = \log(M_{\mathrm{d}}/M_{\mathrm{c}})$ gives $M_{\mathrm{d}} - M_{\mathrm{c}} = M_{\mathrm{c}}(e^{\ell}-1) \geq M_{\mathrm{c}}\ell$. Integration over $1 \leq \tau \leq 2$ yields

$$J(a,b;c,e) \geq c_3(b^2-a^2). \tag{4.3}$$

For the second bound let $\rho = (a+b)/\sqrt{2}$ be large and $0 < d = (b-a)/\sqrt{2} < 1$. Consider points in the strip

$$|\rho' - \rho| \le 1, \qquad 1 \le d' \le 2, \qquad \rho' = (c+e)/\sqrt{2}, \quad d' = (e-c)/\sqrt{2}.$$

All four radii are comparable to ρ. The squared cross distance in radial variables is

$$(a-e)^2 + (b-c)^2 = (\rho - \rho')^2 + (d+d')^2 \le 10.$$

In the rectangular integral (4.2), $2\tau uv \ge 1$ for large ρ. Since $(\log B)'$ increases and is positive at 1, the integrand has a positive lower bound. The side lengths $b-a = \sqrt{2}d$ and $e-c = \sqrt{2}d' \ge \sqrt{2}$ thus give a lower bound $c_4 d$ for the logarithmic ratio. For $z \ge 1$, restrict the spherical integral defining $B(z)$ to $1 - z^{-1} \le \omega_1 \le 1$, where $e^{z\omega_1} \ge e^{z-1}$. The coordinate density from Section 3 is proportional to $(1-\omega_1^2)^{1/2}$ on S^3. On this interval it is bounded below by a constant times $(1-\omega_1)^{1/2}$, whose integral is a constant times $z^{-3/2}$. It follows that $B(z) \ge c_5 e^z z^{-3/2}$. Using this bound in the two factors of M_{c} gives

$$M_{\mathrm{c}} \ge c_6 \rho^{-6} e^{-\tau((a-e)^2+(b-c)^2)} \ge c_7 \rho^{-6} \qquad (1 \le \tau \le 2).$$

Hence $J \ge c_8 d\, \rho^{-6}$ on the strip. Since $c^3 e^3 \asymp \rho^6$ and the change of variables $(c,e) \mapsto (\rho', d')$ is orthogonal, integration gives

$$\int_{|\rho'-\rho|\le 1,\ 1\le d'\le 2} J(a,b;c,e)\, d\mu \ge c_9 d. \tag{4.4}$$

For every $X \in D$, the multiplication coefficient satisfies

$$A(X) \ge c_{10} d^{-\sigma}. \tag{4.5}$$

Indeed, keep the block directions and swap the radii of X. If one block vanishes, choose its direction arbitrarily. The resulting point $X^\sharp$ has signed distance $-d$. By the matching-directions distance formula in Section 3, $|X - X^\sharp| = \sqrt{2}(b-a) = 2d$. The ball $B_{d/2}(X^\sharp)$ lies in D^c by the Lipschitz bound for d_Σ. On this ball $|X - Y| \le 5d/2$, so integrating $K_8(X-Y)$ there gives at least $c d^8 d^{-8-\sigma}$.

Comparison on fixed balls. For every $R > 0$ we claim

$$S(X) \ge c_R(b^2 - a^2) \qquad (X \in D \cap B_R). \tag{4.6}$$

Let $\phi(X) = (|y|^2 - |x|^2)\chi(|X|^2)$, with χ smooth and nonnegative, equal to one on $[0, R^2]$ and vanishing for $r \ge (R+1)^2$. Choose a smooth bump ψ invariant under both block rotations, nonnegative on D and supported on a compact subset of $D \setminus \overline{B}_{R+2}$ away from the factor axes. Take it positive on a closed rectangle $\mathcal{Q}$ of orbit radii $0 < c < e$ with nonempty interior. Extend it by oddness. At $X \in B_{R+1} \cap D$, $\psi(X) = 0$, so (4.1) and (4.3) give

$$L_{\mathrm{o}}\psi(X) = -\int_{c<e} \psi(Y) J(X,Y)\, d\mu(Y) \le -c(b^2-a^2) \int_{\mathcal{Q}} \psi(Y)\, d\mu(Y).$$

The last integral is strictly positive. Also $L\phi$ is smooth, invariant and odd, by differentiation under the absolutely convergent increment integral. For any smooth invariant odd G one has

$$|G(ae_1, be_1)| \le C_R(b^2 - a^2), \qquad 0 \le a \le b \le R+1.$$

In fact, $G(ae_1, ae_1) = 0$, and the derivative in the second radius vanishes at zero by evenness. Its magnitude on the compact square is bounded by a constant times the second radius, using the second derivative bound. Integration from a to b gives the claim.

Apply this bound to $L\phi$ and choose M_0 large enough that $L_{\mathrm{o}}(\phi+M_0\psi) \le 0$ on $B_{R+1} \cap D$. For $\varepsilon > 0$ small, $w = \varepsilon(\phi + M_0\psi)$ lies below S outside that region in D, by strict positivity on the compact support of ψ there. Both functions vanish at the cone. If $w - S$ were positive anywhere in D, it would attain its positive maximum on $\overline{D} \cap \overline{B}_{R+1}$, and the boundary inequalities would put the maximum in $B_{R+1} \cap D$. The nonnegative J and positive A would imply $L_{\mathrm{o}}(w - S) > 0$ at the maximum, while $Lw \le 0 \le f(S) = LS$ there. This proves (4.6).

The dilation derivative $P(X) = X \cdot \nabla S(X)$ is smooth, block invariant and odd, since S has these properties and dilations commute with the block symmetries. In the next estimate the second derivatives of P on a compact ball are controlled by the derivatives of S through order three, supplied by Lemma 2. The same bound for $G = P$ on B_{2R} gives $|P(\ell X)| \le C_{2R}\ell^2(b^2-a^2)$ when $X \in B_R \cap D$ and $1 \le \ell \le 2$. Integrate $\frac{d}{d\ell}S(\ell X) = \ell^{-1}P(\ell X)$ from 1 to $\lambda \le 2$, then use (4.6) to obtain

$$S(\lambda X) - S(X) \le C_R(\lambda - 1)S(X) \qquad (X \in D \cap B_R). \tag{4.7}$$

A strict linearized inequality outside a ball. Fix $0 < \vartheta < 1$ and set $F(u) = u - \vartheta u^3$. Then $F(S)$ is odd under T, and $(1-\vartheta)S \le F(S) \le S$ on D. For $u, v \ge 0$,

$$F(u) - F(v) - F'(u)(u - v) = \vartheta(v - u)^2(v + 2u).$$

Subtract $F'(S(X))L_{\mathrm{o}}S(X)$ from $L_{\mathrm{o}}(F(S))(X)$ in (4.1). The integral part becomes the displayed cubic remainder with $u = S(X)$, $v = S(Y)$. The multiplication part becomes $A(X)(F(S) - F'(S)S) = 2\vartheta A(X)S^3$, with $S = S(X)$. Since $L_{\mathrm{o}}S = f(S)$ and $F'(S)f(S) - f'(S)F(S) = 2(1-\vartheta)S^3$, we obtain

$$\begin{aligned}(L_{\mathrm{o}} - f'(S))F(S)(X) &= 2(1-\vartheta)S(X)^3 + 2\vartheta A(X)S(X)^3 \\ &\quad + \vartheta \int_{c<e} (S(Y) - S(X))^2 \\ &\qquad \times (S(Y) + 2S(X))J(X,Y)\, d\mu(Y).\end{aligned} \tag{4.8}$$

For large ρ, the saddle asymptotics give $S(Y) \ge q(1)/2 =: m_0 > 0$ on the strip in (4.4). Also $S(X) \le Cd$, since S vanishes on Σ and has bounded gradient. Choose $d_0 \in (0,1)$ small enough that $S(X) \le m_0/2$ for $d \le d_0$. Then on the strip the product $(S(Y) - S(X))^2(S(Y) + 2S(X))$ is at least

$m_0^3/4$. By (4.4), the integral in (4.8) is bounded below by a positive constant times d, and hence by a positive constant times $S(X)$. When $d \geq d_0$, the positive lower bound for S away from the cone gives $2(1-\vartheta)S^3 \geq cS$. All terms in (4.8) are nonnegative, so these two ranges give constants $c_0 > 0$ and $R_0 < \infty$ with

$$(L_{\mathrm{o}} - f'(S))F(S) \geq c_0 S \qquad (X \in D \setminus B_{R_0}). \tag{4.9}$$

The dilation comparison. Put $S_\lambda(X) = S(\lambda X)$ and $\delta = \lambda - 1 > 0$. Changing variables in the kernel integral gives $LS_\lambda(X) = \lambda^\sigma LS(\lambda X)$, so $LS_\lambda = \lambda^\sigma f(S_\lambda)$. Let $V = S + C\delta F(S)$. With $C\delta \leq 1$, we have $f(V) \leq V \leq 2S$ on D. Concavity of f on $[0,\infty)$ and $\lambda^\sigma - 1 \leq \sigma\delta$ give, by (4.9),

$$\begin{aligned} L_{\mathrm{o}}V - \lambda^\sigma f(V) &\geq C\delta(L_{\mathrm{o}} - f'(S))F(S) - (\lambda^\sigma - 1)f(V) \\ &\geq \delta(Cc_0 - 2\sigma)S \qquad (X \in D \setminus B_{R_0}). \end{aligned}$$

For $1 < \lambda \leq 2$, choose C large enough for this to be positive and for $S_\lambda \leq V$ on $D \cap \overline{B}_{R_0}$, using (4.7), continuity, and $F(S) \geq (1-\vartheta)S$. The choice of C is independent of δ.

The comparison at infinity uses the relative layer asymptotics

$$\sup_{\substack{X \in D \\ |X| > R}} \left| \frac{S(X)}{q(d_\Sigma(X))} - 1 \right| \longrightarrow 0 \qquad (R \to \infty). \tag{4.10}$$

For $d \geq 1$, this follows from the uniform value asymptotics and $q(d) \geq q(1) > 0$. For $0 < d \leq 1$ and large $|X|$, join the cone to X by keeping ρ and both block directions fixed and varying the signed distance from 0 to d. This is a segment of length d, with radii $(\rho - r)/\sqrt{2}$ and $(\rho + r)/\sqrt{2}$ at distance r. It stays off the factor axes and at distance at least $\rho \geq |X|/\sqrt{2}$ from the origin. At the cone endpoint $S = 0$ and $q(d_\Sigma) = q(0) = 0$ by the centering in Section 2. Integrating their gradient difference bounds the value difference by d times a quantity tending uniformly to zero. On $[0,1]$, the continuous positive derivative q' has a positive minimum. Thus $q(d) \geq cd$ there, which gives (4.10) also for $0 < d \leq 1$.

The concavity of the same layer on $(0,\infty)$ follows from [4, Lemma 4.2, at extension height zero]. For $d > 0$ and $\lambda \geq 1$, write $d = \lambda^{-1}(\lambda d) + (1-\lambda^{-1})0$. Concavity and $q(0) = 0$ give $q(\lambda d) \leq \lambda q(d)$.

For each $0 < \delta \leq 1$, choose an outer comparison radius greater than R_0 so that the relative error in (4.10) is at most $\delta/4$ outside that radius. If X is outside, so is λX, whose cone distance is λd. Therefore

$$\frac{S_\lambda(X)}{S(X)} \leq \frac{1+\delta/4}{1-\delta/4}\frac{q(\lambda d)}{q(d)} \leq \frac{(1+\delta/4)(1+\delta)}{1-\delta/4} \leq 1 + 3\delta.$$

Choose $C(1-\vartheta) \geq 3$, in addition to the earlier conditions on C, and then restrict δ by $C\delta \leq 1$. This gives $S_\lambda \leq V$ on the outer region as well as on the inner ball. Continuity gives the comparison on the sphere of the outer radius as well.

To compare in the remaining annulus write $h = S_\lambda - V$ and $\mathcal{A}_\lambda = L_{\mathrm{o}} - \lambda^\sigma f'(S)$. At a point where $h > 0$ in the exterior of B_{R_0}, concavity and $V \geq S$ imply

$$f(S_\lambda) - f(V) \leq f'(V)h \leq f'(S)h, \qquad \mathcal{A}_\lambda h \leq 0.$$

Since $f'(S) \leq 1$ and $0 \leq F(S) \leq S$ in D, (4.9) gives

$$\mathcal{A}_\lambda F(S) \geq \big(c_0 - (\lambda^\sigma - 1)\big)S \geq \tfrac{c_0}{2}S \qquad \text{in } D \setminus B_{R_0}$$

for sufficiently small δ. Use the positive divisor $\Psi = F(S) + \eta$ on D, extended oddly, for a constant $\eta > 0$ to be chosen. Its jump at the cone is a positive distance from each evaluation point in D, so the operator there is well defined. The constant η on D drops out of the first term of (4.1) and contributes $\eta(A - \lambda^\sigma f'(S))$ to $\mathcal{A}_\lambda\Psi$. Choose $d_* \in (0,1)$ so that $c_{10}d_*^{-\sigma} \geq 2$. For $d \leq d_*$, (4.5) and $\lambda^\sigma \leq 2$ imply that this contribution is nonnegative. For $d \geq d_*$, use $m_* = \inf_{\{d_\Sigma \geq d_*\}} S > 0$. This lower bound holds with the original R_0 used to choose C. Since $A - \lambda^\sigma f'(S) \geq -2$, choosing $0 < \eta < c_0 m_*/8$ gives $\mathcal{A}_\lambda\Psi > 0$ throughout the exterior. If h were positive somewhere, h/Ψ would attain a positive maximum m inside $D \setminus B_{R_0}$. Indeed, the quotient is nonpositive on the inner ball and on the outer region, including the two radial boundary spheres. On the cone portion of its boundary the quotient extends continuously by zero because $h = 0$ there and $\Psi \geq \eta$ on D. A positive maximum therefore occurs at an interior point X_* in D outside the inner ball. The function $h - m\Psi$ is nonpositive on D and vanishes at X_*. At this point the multiplication terms in $\mathcal{A}_\lambda(h - m\Psi)$ vanish because $(h - m\Psi)(X_*) = 0$. Thus

$$\mathcal{A}_\lambda(h - m\Psi)(X_*) = -\int_{c<e} (h - m\Psi)(Y)J(X_*, Y)\, d\mu(Y) \geq 0.$$

Since $m > 0$ and $\mathcal{A}_\lambda\Psi(X_*) > 0$, this implies $\mathcal{A}_\lambda h(X_*) > 0$, a contradiction.

Off the axes in D the radial monotonicities give $P = \rho S_\rho + dS_d \geq 0$, and continuity extends this to the axes. Integration along rays thus gives $S_\lambda \geq S$ in D. The upper comparison just proved bounds $(S_\lambda - S)/\delta$ on D for all sufficiently small $\delta > 0$. Oddness gives a global absolute bound. Letting $\delta \downarrow 0$ gives $P \in L^\infty(\mathbb{R}^8)$. On D it follows that $\rho S_\rho \leq P$, and oddness and continuity extend the bound for $|S_\rho|$ to $|d| < \rho$.

It remains to bound ∇P. Take the difference quotient of the scaled equation in distributions. For a compactly supported smooth test function φ, Lemma 2 gives $L\varphi \in L^1$. The global quotient bound therefore permits dominated convergence against $L\varphi$. On the source side write

$$\frac{\lambda^\sigma f(S_\lambda) - f(S)}{\delta} = \lambda^\sigma \frac{f(S_\lambda) - f(S)}{\delta} + \frac{\lambda^\sigma - 1}{\delta} f(S).$$

Both terms are uniformly bounded by the quotient estimate and smoothness of the cubic on $[-1,1]$. They converge pointwise to $f'(S)P$ and $\sigma f(S)$, respectively. This gives $LP = f'(S)P + \sigma f(S)$ in distributions.

The linear estimates of Lemma 2 apply to this distributional equation with bounded P and bounded source. They first give C^σ bounds on smaller balls. The coefficients $f'(S)$ and $f(S)$ have uniform bounded Lipschitz norms, so the source has a C^α bound in terms of that of P whenever $0 < \alpha < 1$. Choose $0 < b < \sigma$ with $jb + \sigma \neq 1$ for every positive integer j, excluding only countably many choices. Starting with C^b, each step with $jb < 1$ gives $C^{jb+\sigma}$ on a smaller ball and hence $C^{(j+1)b}$ if $jb + \sigma < 1$. At the first j with $jb + \sigma > 1$, one still has $jb < 1$ because $b < \sigma$. This finite iteration on fixed nested radii inside each translated unit ball gives a C^1 bound independent of the center. Thus ∇P is bounded globally.

Finally, at fixed ρ and fixed block directions ω, ω', varying d traces the affine line

$$X = \left(\frac{\rho - d}{\sqrt{2}}\omega, \frac{\rho + d}{\sqrt{2}}\omega' \right).$$

It has unit velocity, so the derivatives in d are ordinary Cartesian directional derivatives on the region with both radii positive. Differentiating $P = \rho S_\rho + dS_d$ gives $\rho S_{\rho d} = P_d - S_d - dS_{dd}$. The bound for ∇P and the bounds for ∇S and $D^2 S$ from Lemma 2 control the right-hand side on each fixed band $|d| \leq M$ with $\rho \geq 2M + 1$. □

5. Transport in the exterior of the cap

We keep the cone coordinates of Section 3, with $a = |x|$ and $b = |y|$. In the exterior tube about the cap the map sends signed normal distance to the cone coordinate t, and arclength along the cap's meridian to the cone coordinate ρ. Its differential at the exterior cap is therefore an isometry in the radial plane. In the six perpendicular directions it dilates the two angular blocks. The corresponding change in the affine kernel has zero integral in those six directions for each fixed nonzero radial displacement. We use this cancellation after restricting the profile to the radial plane.

Proposition 6. *Use the exponent and operator conventions of Section 2 and the curvature convention of Section 3. Let v be a smooth solution of $Lv = f(v)$ on $\mathbb{R}^8$, with values in $(-1, 1)$, that depends only on a, b, is nonincreasing in a and nondecreasing in b, and converges uniformly to the corresponding well as $t \to \pm\infty$.*

Let $E_1 \subset D$ be a smooth cap invariant under the block rotations. Suppose that outside a compact set membership in E_1 is equivalent to $t > \eta_1(\rho)$, where $\eta_1 > 0$ is smooth and, for some $0 < \kappa < 7$, satisfies

$$\eta_1(q) \asymp q^{-\kappa}, \qquad |\eta_1^{(j)}(q)| \leq C q^{-\kappa - j} \quad (1 \leq j \leq 3).$$

For $\lambda \geq 1$ put $E_\lambda = \lambda E_1$, $\eta_\lambda(q) = \lambda \eta_1(q/\lambda)$, and

$$m_\lambda(r) = \lambda^{\kappa+1} r^{-\kappa-1-\sigma}.$$

There exist $L_0 > 1$ and smooth, uniformly bi-Lipschitz diffeomorphisms $\Phi_\lambda = \lambda \Phi_1(\cdot/\lambda)$ commuting with the block rotations, with the following properties for every fixed $L_1 \geq L_0$.

(1) *The function* $u_\lambda = v \circ \Phi_\lambda$ *satisfies* $u_\lambda \le v$, *and the displacement satisfies*

$$|\Phi_\lambda(P) - P| \le C\lambda^{\kappa+1}(\lambda + |P|)^{-\kappa}.$$

The sets $F_\lambda = \Phi_\lambda^{-1}(D)$ *and* E_λ *coincide outside* $B_{L_1\lambda}$. *There is* $c > 0$ *such that at* $P = p + d\nu(p)$, *with* $p \in \partial E_\lambda$, ν *pointing into the cap,* $|P| \ge L_1\lambda$, *and* $|d| \le c|p|$, *the map preserves the two block directions and has radial coordinates*

$$(\rho, t)\big(\Phi_\lambda(P)\big) = \big(S_\lambda(\rho(p)), d\big), \qquad S_\lambda(q) = q - \int_q^\infty \big(\sqrt{1 + \eta_\lambda'(u)^2} - 1\big)\, du.$$

The arclength label satisfies $S_\lambda(q) \asymp q$ *uniformly for* $q \ge L_0\lambda/\sqrt{2}$ *after increasing* L_0 *if needed. On the tube* Φ_λ *strictly increases* a *and strictly decreases* b.

(2) *If* $V_\lambda : \mathbb{R}^8 \to [-1, 1]$ *is measurable and equals* u_λ *outside* $B_{L_1\lambda}$, *then*

$$|(LV_\lambda - f(V_\lambda))(P)| \le C_{L_1} m_\lambda(|P|), \qquad |P| \ge 2L_1\lambda.$$

(3) *Suppose also that*

$$\lambda^{-8} \int_{B_{L_1\lambda}} |V_\lambda - (2\chi_{E_\lambda} - 1)|\, dQ \longrightarrow 0.$$

For every finite T, *as* $\lambda \to \infty$,

$$(LV_\lambda - f(V_\lambda))(P) = c_{8,s} H_\sigma[E_\lambda](p) + o\big(m_\lambda(|P|)\big) \tag{5.1}$$

uniformly at $P = p + d\nu(p)$ *with* $|d| \le T$ *and* $|P| \ge 2L_1\lambda$. *In particular, if* $H_\sigma[E_1](p) \le -c(1 + |p|)^{-\kappa-1-\sigma}$ *on* ∂E_1, *the residual in* (5.1) *is at most* $-c_* m_\lambda(|P|)$ *for a constant* $c_* > 0$ *independent of* T *once* λ *is sufficiently large depending on* T.

Proof. The end-state hypothesis gives agreement of the signs of v and t for large $|t|$. Thus Lemma 4, with the constant enlarged for bounded $|t|$, gives

$$|v(X) - (2\chi_D(X) - 1)| + |\nabla v(X)| \le C(1 + |t(X)|)^{-\sigma}. \tag{5.2}$$

The map. We work first in the plane of block radii. Put

$$e_\lambda(R) = \lambda^{\kappa+1}(\lambda + R)^{-\kappa}, \qquad \delta_\lambda(R) = \frac{e_\lambda(R)}{\lambda + R}.$$

For q beyond a fixed multiple of λ, the scaled height bounds give

$$\eta_\lambda(q) \asymp e_\lambda(q), \qquad |\eta_\lambda^{(j)}(q)| \le C e_\lambda(q) q^{-j} \quad (1 \le j \le 3).$$

These estimates hold with $e_\lambda(\rho)$ in place of $e_\lambda(q)$ whenever $\rho/2 \le q \le 2\rho$. The ratio $e_\lambda(\rho)/\rho$ tends uniformly to zero as ρ/λ grows.

Choose $0 < c_0 \le 1/8$ small and then $C_0 > 1$ large. Take $C_0/2$ beyond the exterior graph threshold for E_1. In $\rho > C_0\lambda$, $|t| < c_0\rho$, the inward normal coordinates, with q the radial label at the footpoint, have the form

$$(\rho, t) = (q, \eta_\lambda(q)) + \frac{d}{\sqrt{1 + \eta_\lambda'(q)^2}}\, (-\eta_\lambda'(q), 1).$$

Eliminating d gives

$$\rho = q - (t - \eta_\lambda(q))\eta_\lambda'(q).$$

For $\rho/2 \le q \le 2\rho$ the right side is $q + O(e_\lambda(\rho))$, so it brackets ρ at the endpoints. Its derivative in q is $1+\eta_\lambda'(q)^2-(t-\eta_\lambda(q))\eta_\lambda''(q)$, which is within $1/2$ of one by the height bounds. Thus q exists uniquely and is smooth in (ρ, t).

A cap point at radial label ρ with the same block directions has distance $|t - \eta_\lambda(\rho)| < \rho/4$, by the choices of c_0, C_0. The intersection of the closed boundary with the closed ball of radius $\rho/4$ centered at the source point is therefore a nonempty compact set, on which the distance attains its minimum. Any boundary point with label outside $(\rho/2, 2\rho)$ is at distance at least $\rho/2$ by the Lipschitz bound for ρ in Section 3. At the remaining radii both blocks are nonzero, so the distance calculation in that section forces matching block directions. Thus the minimizing point lies on the exterior graph in the open interval $(\rho/2, 2\rho)$, where stationarity of the squared distance gives the normal equation. The uniqueness just proved identifies its label with q, so $|d|$ is the distance to the boundary. Its sign agrees with $t - \eta_\lambda(\rho)$ because

$$t - \eta_\lambda(\rho) = \frac{d}{\sqrt{1+\eta_\lambda'(q)^2}} + \eta_\lambda(q) - \eta_\lambda(\rho).$$

Since $|q - \rho| \le |d||\eta_\lambda'(q)|$, the mean value theorem gives

$$|\eta_\lambda(q) - \eta_\lambda(\rho)| \le |d| \sup_{[\rho/2, 2\rho]} |\eta_\lambda'|^2.$$

For C_0 large this is smaller than the magnitude of the first term whenever $d \neq 0$. This proves that d is the signed distance, positive in the cap.

The exact map sends (ρ, t) to $(S_\lambda(q), d)$. We first check its displacement estimates in the meridional plane. Write $F(q,t) = (t - \eta_\lambda(q))\eta_\lambda'(q)$, so $q = \rho + F(q,t)$. In this region $|F| \le Ce_\lambda(\rho)$, $|DF| \le Ce_\lambda(\rho)/\rho$, and $|D^2F| \le Ce_\lambda(\rho)/\rho^2$, with the derivatives taken in q, t. For example,

$$F_q = (t-\eta_\lambda)\eta_\lambda'' - (\eta_\lambda')^2, \qquad F_{qq} = (t-\eta_\lambda)\eta_\lambda^{(3)} - 3\eta_\lambda'\eta_\lambda''.$$

Implicit differentiation gives $(1-F_q)Dq = D\rho + F_t Dt$, so Dq is bounded since $1 - F_q \ge 1/2$. The identity $D(q-\rho) = F_q Dq + F_t Dt$ then gives $|D(q-\rho)| \le Ce_\lambda(\rho)/\rho$. Differentiating $q - \rho = F(q,t)$ twice leaves $(1-F_q)D^2q$ on the left and terms containing D^2F and the bounded first derivatives of q, t on the right. This gives $|D^2q| \le Ce_\lambda(\rho)/\rho^2$. For $S_\lambda(q) - q$, integrating the bound $\sqrt{1+(\eta_\lambda')^2} - 1 \le C(\eta_\lambda')^2$ from q to infinity gives $O(e_\lambda(q)^2/q)$. In particular, for q/λ large enough, $0 \le q - S_\lambda(q) \le q/2$, giving the stated comparability. The defining integral gives $S_\lambda'(q) = \sqrt{1+\eta_\lambda'(q)^2}$, so S_λ is arclength up to an additive constant. The first and second derivatives of $S_\lambda(q) - q$ are $O(e_\lambda(q)^2/q^2)$ and $O(e_\lambda(q)^2/q^3)$, respectively. For the other

component, write

$$d - t = -\eta_\lambda(q) + (t - \eta_\lambda(q))\big(\sqrt{1+\eta_\lambda'(q)^2} - 1\big).$$

The height derivatives through order three give

$$\left|\partial_q^j\big(\sqrt{1+\eta_\lambda'(q)^2} - 1\big)\right| \le Ce_\lambda(q)^2 q^{-2-j}, \qquad j = 0,1,2.$$

Since $|t - \eta_\lambda(q)| \le C\rho$, the product rule and the bounds on Dq, D^2q give

$$|D^j(d-t)| \le Ce_\lambda(\rho)\rho^{-j}, \qquad j = 0,1,2.$$

The bounds on $q - \rho$ and $S_\lambda(q) - q$ give the same orders for $S_\lambda(q) - \rho$.

To extend the map, fix $0 < \tau < c_0/4$. Multiply the displacement in (ρ, t) by a product of smooth cutoffs with values in $[0,1]$. Use a radial cutoff supported in $\rho > C_0\lambda$ and equal to one for $\rho \ge 2C_0\lambda$, and an angular cutoff supported in $|t| < c_0\rho$ and equal to one for $|t| \le 2\tau\rho$. The cutoffs depend only on $\rho/(C_0\lambda)$ and t/ρ, respectively, and their product vanishes near the boundary of the normal-coordinate region. Lift the map by retaining both block directions. On its support $a, b \asymp \rho$, and the target radii are positive since the displacement is $o(\rho)$ as ρ/λ grows. The map is the identity off the support, including near either axis. Each derivative of a cutoff costs one inverse power of ρ. For the radial cutoff this uses $\rho \asymp C_0\lambda$ on its transition region, and for the angular cutoff it follows by differentiating t/ρ. If the first radius changes by Δa, the first Cartesian block changes by $\Delta a\, x/|x|$. Since $|x| \asymp \rho$ on the support, the first two derivatives of the direction have bounds $C\rho^{-1}$ and $C\rho^{-2}$, and similarly for the second block. The Cartesian displacement and its first two derivatives therefore obey

$$\begin{aligned} |\Phi_\lambda(Q) - Q| &\le Ce_\lambda(|Q|), \\ |D\Phi_\lambda(Q) - I| &\le C\delta_\lambda(|Q|), \\ |D^2\Phi_\lambda(Q)| &\le Ce_\lambda(|Q|)(\lambda + |Q|)^{-2}. \end{aligned} \tag{5.3}$$

Increase C_0 with c_0, τ fixed so that the displacement has Lipschitz constant less than $1/4$. For each prescribed target Y, the map $Q \mapsto Y - (\Phi_\lambda(Q) - Q)$ is a contraction on $\mathbb{R}^8$, whose fixed point is the unique preimage of Y. The Lipschitz bound also gives the uniform inverse Lipschitz bound, and the inverse function theorem gives smoothness. The arclength integral gives $S_\lambda(q) = \lambda S_1(q/\lambda)$ by substitution. Together with the scaling of the height, normal coordinates and cutoffs this yields $\Phi_\lambda = \lambda\Phi_1(\cdot/\lambda)$.

Before cutting off, the change of t is $-\eta_\lambda(q) + d(1 - 1/\sqrt{1+\eta_\lambda'(q)^2})$. Here $|d| \le C(c_0 q + e_\lambda(q))$ and $|\eta_\lambda'| \le Ce_\lambda(q)/q$. Thus the second term is at most $\eta_\lambda(q)/4$ in absolute value if C_0 is large. The change of ρ is at most $|S_\lambda(q) - q| + |d\eta_\lambda'(q)|$ in absolute value. Dividing this by $\eta_\lambda(q) \asymp e_\lambda(q)$ bounds the ratio by $C(c_0 + e_\lambda(q)/q)$, which is at most $1/2$ for c_0 small and C_0 large. The change of t is therefore negative and dominates the change of ρ. As $a = (\rho - t)/\sqrt{2}$ and $b = (\rho + t)/\sqrt{2}$, the exact map increases a

and decreases b. The cutoffs preserve the corresponding weak inequalities, which give $u_\lambda \le v$.

Take $L_0 \ge 2\sqrt{2}C_0$, so that at $|P| \ge L_0\lambda$ the radial cutoff is one by $\rho(P) \ge |P|/\sqrt{2}$. In the angular region $|t| \le 2\tau\rho$, the sign after transport is the sign of d, hence the cap sign. In its complement, $|t| > 2\tau\rho$. Choose L_0 large enough that $\eta_\lambda(\rho)$ and the change in t are at most $\tau\rho/4$ in absolute value throughout the exterior. The cap sign and transported cone sign then both agree with the original cone sign. This proves $F_\lambda = E_\lambda$ on the exterior.

For the tube assertion, fix $0 < c \le \min\{1/8, \tau/2\}$. If $P = p + d\nu(p)$, $|P| \ge L_1\lambda$, and $|d| \le c|p|$, then $|p| \ge L_0\lambda/(1+c)$. Increase L_0 so that these p belong to the exterior graph and $\eta_\lambda(\rho(p)) \le \tau|p|/2$. The block rotation directions are tangent to the graph, so $\nu(p)$ belongs to the radial plane. Both block radii at p then exceed $|p|/4$, by the inverse formulas in Section 3. Hence the normal segment stays in the same radial plane without reaching either axis. The Lipschitz bounds give

$$\rho(P) \ge (1/\sqrt{2} - c)|p| \ge |p|/2, \qquad |t(P)| \le (\tau/2 + c)|p| \le 2\tau\rho(P).$$

Together with $|\rho(P) - \rho(p)| \le c|p|$, these give $\rho(p) \in [\rho(P)/2, 2\rho(P)]$ and put P where both cutoffs are one. The given normal segment satisfies the planar normal equation with $q = \rho(p)$, so uniqueness in this interval identifies p as the footpoint used by the exact map.

Kernel estimates. In the calculations below suppress the index λ, put $K(h) = |h|^{-8-\sigma}$, and write $J(Q) = \det D\Phi(Q)$. The determinant $J(Q)$ is positive because $\|D\Phi - I\| < 1/4$. Use $f(u(P)) = (Lv)(\Phi(P))$ and change variables $Y = \Phi(Q)$ in the latter expression to obtain

$$\begin{aligned} c_{8,s}^{-1}(Lu - f(u))(P) &= \int_{\mathbb{R}^8} (u(P) - u(Q))A_P(Q)\,dQ, \\ A_P(Q) &= K(Q-P) - J(Q)K(\Phi(Q) - \Phi(P)). \end{aligned} \tag{5.4}$$

Both integrals before subtraction converge absolutely. Near the diagonal this follows from the Lipschitz bound for v in (5.2), the bi-Lipschitz bounds for Φ, and $\sigma < 1$. Away from the diagonal it follows from boundedness.

Two consequences of (5.2) will be used. For any set G write $w_G = 2\chi_G - 1$, and put $D_P = |t(\Phi(P))|$. If D_P is large, then $t \circ \Phi$ has the same sign as at P, and has magnitude at least $D_P/2$, throughout $B_{cD_P}(P)$ for a fixed small $c > 0$. This uses the Lipschitz bounds of t and Φ. In this ball, the bound for ∇v and the chain rule control the increments of u by $CD_P^{-\sigma}|h|$ for $|h| \le 1$, and the well bound controls them by $CD_P^{-\sigma}$ for $1 < |h| \le cD_P$. The remaining kernel tail is $O(D_P^{-\sigma})$. For bounded D_P use the global Lipschitz bound and boundedness. Consequently

$$\int_{\mathbb{R}^8} |u(P) - u(P+h)|K(h)\,dh \le C(1 + D_P)^{-\sigma}. \tag{5.5}$$

Also, for $R \geq 1$,

$$\int_{B_R} |u - w_F| \, dQ \leq CR^{8-\sigma}. \tag{5.6}$$

Indeed $\Phi(0) = 0$ and $\Phi(B_R) \subset B_{CR}$ by the uniform Lipschitz bound. Change variables to $\Phi(Q)$, using bounded distortion. The absolute difference is at most $C(1+|t|)^{-\sigma}$ by (5.2). In block polar coordinates the volume density is proportional to $a^3b^3 \, d\rho \, dt$, bounded on B_{CR} by $CR^6 \, d\rho \, dt$. The ρ-range has length $O(R)$ and $\int_{-CR}^{CR}(1+|t|)^{-\sigma} \, dt = O(R^{1-\sigma})$, giving (5.6).

For the exterior kernel estimates take $r = |P| \geq 3L_1\lambda/2$. Abbreviate $e = e_\lambda(r)$, $\delta = \delta_\lambda(r)$, and $m = m_\lambda(r)$, so $\delta r^{-\sigma} \asymp m$. Take $0 < \theta \leq \theta_0 < 1/6$, with θ_0 to be reduced below. The inequality $r \geq 3L_1\lambda/2$ gives $B_{2\theta r}(P) \subset \{|Q| > L_1\lambda\}$, and $|Q| \asymp r$ on this ball.

Write $h = Q - P$ and $g = \Phi(Q) - \Phi(P)$. The displacement has Lipschitz constant less than $1/4$, so $|g - h| \leq |h|/4$ and the entire segment from h to g stays away from the singularity by at least $3|h|/4$. Also

$$|g - h| \leq C(e_\lambda(r) + e_\lambda(|Q|)), \qquad |J(Q) - 1| \leq C\delta_\lambda(|Q|).$$

The mean value inequality for K then gives, on $|Q - P| > \theta r$,

$$|A_P(Q)| \leq CK(Q-P)\left(\delta_\lambda(|Q|) + \frac{e_\lambda(r) + e_\lambda(|Q|)}{|Q-P|}\right). \tag{5.7}$$

For $0 < b_0 < 1/2$, the part of the integral of $|A_P|$ with $|Q| < b_0 r$ is bounded by $Cb_0^{7-\kappa}m$. In fact $K(Q-P) \leq Cr^{-8-\sigma}$ there, and the radial integrals of $\delta_\lambda(|Q|)$ and $e_\lambda(|Q|)/r$ are bounded by

$$C\lambda^{\kappa+1}(b_0 r)^{7-\kappa}, \qquad C\lambda^{\kappa+1}(b_0 r)^{8-\kappa}/r,$$

respectively, by integrating the bounds $\delta_\lambda(R) \leq \lambda^{\kappa+1}R^{-\kappa-1}$ and $e_\lambda(R) \leq \lambda^{\kappa+1}R^{-\kappa}$ against $CR^7 \, dR$. The condition $\kappa < 7$ ensures convergence at zero. The remaining $e_\lambda(r)/r$ term contributes $Cb_0^8 m$. For $B_0 > 2$, the integral with $|Q| > B_0 r$ is at most $CB_0^{-\sigma}m$, since the factor in parentheses in (5.7) is bounded by $C\delta$ there. On $b_0 r \leq |Q| \leq B_0 r$ outside $B_{\theta r}(P)$, the parenthesis in (5.7) is at most $C_{\theta,b_0}\delta$, and $K(Q-P) \leq C_\theta r^{-8-\sigma}$. Hence $|A_P(Q)| \leq C_{\theta,b_0,B_0} m r^{-8}$. In particular, fixing b_0 and B_0 and using all three regions gives $\int_{|Q-P|>\theta r} |A_P(Q)| \, dQ \leq C_\theta m$. We also need an integrated derivative bound for shifts of the center. Differentiating the two kernels in P gives

$$\nabla_P A_P(Q) = -\nabla K(Q-P) + J(Q)D\Phi(P)^{\mathsf{T}}\nabla K(\Phi(Q) - \Phi(P)).$$

The matrix in the second term differs from the identity by

$$\|J(Q)D\Phi(P)^{\mathsf{T}} - I\| \leq C\big(\delta_\lambda(|Q|) + \delta_\lambda(r)\big).$$

Together with $|D^2 K| \leq CK(h)/|h|^2$ along the segment from h to g, this gives

$$|\nabla_P A_P(Q)| \leq \frac{CK(h)}{|h|}\left(\delta_\lambda(|Q|) + \delta_\lambda(r) + \frac{e_\lambda(r) + e_\lambda(|Q|)}{|h|}\right).$$

On $|h| > \theta r$ the factor $1/|h|$ is at most $1/(\theta r)$. The term with $\delta_\lambda(r)$ integrates to at most $C_\theta \delta_\lambda(r) r^{-1-\sigma}$, and the other terms are bounded by the same split as above. We obtain

$$\int_{|Q-P|>\theta r} |\nabla_P A_P(Q)|\, dQ \le C_\theta m/r.$$

Suppose first that $|t(P)| \le \tau\rho(P)$. The functions t and ρ are 1-Lipschitz, and $\rho(P) \ge r/\sqrt{2}$. For $|Q-P| \le 2\theta r$, we have $|t(Q)| \le \tau\rho(P) + 2\theta r$ and $\rho(Q) \ge \rho(P) - 2\theta r$. Thus $\theta_0 < \tau/[2\sqrt{2}(1+2\tau)]$ keeps the ball in $|t| \le 2\tau\rho$, where the transport is exact. Let $\gamma(q) = (q, \eta_\lambda(q))$, let T_q, N_q be its unit tangent in the direction of increasing q and inward unit normal, and let $k = \eta_\lambda''(1+\eta_\lambda'^2)^{-3/2}$ be its signed curvature. The derivative of N_q in arclength is $-kT_q$, by differentiating the unit normal to the graph. Thus the derivatives of $\gamma(q)+dN_q$ with respect to arclength and to d are $(1-dk)T_q$ and N_q. Since S_λ is the arclength label, the corresponding target derivatives are e_ρ and e_t, the unit vectors of the radial plane. Consequently $D\Phi(P)$ sends T_q to $(1-dk)^{-1}e_\rho$ and N_q to e_t. On the six directions perpendicular to the radial plane, $D\Phi(P)$ multiplies each three-dimensional block by the ratio of the target and source radii. Indeed the radius is constant to first order in an angular direction, and the transport preserves the angular coordinate, so the length of that variation scales by this ratio. Denote these two dilations, extended by the identity on the radial plane, by Λ_P. If O_P is the rotation sending (T_q, N_q) to (e_ρ, e_t) and fixing the perpendicular directions, then

$$\|D\Phi(P) - O_P\Lambda_P\| \le C\delta|d|/r, \qquad \|\Lambda_P - I\| \le C\delta.$$

Here $|k| \le Ce/r^2$ and the exact region has $|d| \le Cr$. Also $\|O_P - I\| \le C|\eta_\lambda'(q)| \le C\delta$, since the meridional rotation has angle $\arctan(\eta_\lambda'(q))$ in absolute value. Write $h = h_\parallel + h_\perp$ for the two orthogonal projections and put

$$A_{\Lambda_P}(h) = K(h) - \det\Lambda_P\, K(\Lambda_P h).$$

The rotation in the radial plane leaves the kernel and the determinant factor invariant. Taylor expansion of Φ and the Jacobian bound therefore give

$$|A_P(P+h) - A_{\Lambda_P}(h)| \le C\delta\,\frac{|h|+|d|}{r}K(h), \qquad |h| < \theta r. \tag{5.8}$$

Indeed $\Phi(P+h) - \Phi(P) - O_P\Lambda_P h = O(\delta(|h|+|d|)|h|/r)$ by Taylor's formula and the displayed differential estimate. The determinant is locally Lipschitz in matrix entries, so $J(P+h) - \det\Lambda_P = O(\delta(|h|+|d|)/r)$. Both kernel arguments, $\Phi(P+h) - \Phi(P)$ and $O_P\Lambda_P h$, differ from h by at most $C\delta|h|$. The first bound follows from (5.3) on the segment from P to $P+h$, and the second from the bounds on O_P and Λ_P. By increasing L_0 we make $C\delta \le 1/4$, so the segment joining the two arguments stays at length comparable to $|h|$, where $|\nabla K| \le C|h|^{-9-\sigma}$. This proves (5.8). The same mean value bound applied to $\Lambda_P - I$ gives $|A_{\Lambda_P}| \le C\delta K$.

For each fixed $h_\parallel \ne 0$, integration of A_{Λ_P} over the perpendicular variables gives zero. Indeed, Λ_P is the identity in the radial plane, so its determinant

is the Jacobian for the change of variables $k_\perp = \Lambda_P h_\perp$ in the six-dimensional perpendicular space. Thus

$$\int_{\mathbb{R}^6} \det \Lambda_P \, K(h_\parallel + \Lambda_P h_\perp)\, dh_\perp = \int_{\mathbb{R}^6} K(h_\parallel + k_\perp)\, dk_\perp.$$

Consequently, any function of $h_\parallel$ whose product with A_{Λ_P} is absolutely integrable has integral zero against A_{Λ_P} by Fubini. An odd function of $h_\parallel$ also has integral zero on each centered spherical truncation whenever that truncated integral is absolutely convergent, since A_{Λ_P} is even. Define $b_P(h_\parallel) = u(P + h_\parallel)$ on $|h_\parallel| < \theta r$ and $b_P = 0$ off that disk. Then

$$\int_{\mathbb{R}^8} \big(u(P) - b_P(h_\parallel)\big) A_{\Lambda_P}(h)\, dh = 0. \tag{5.9}$$

Indeed, $u(P) - b_P(h_\parallel)$ is $O(|h_\parallel|)$ at zero and bounded elsewhere, so this is the absolutely integrable case. The cancellation in (5.9) is taken over all of $\mathbb{R}^8$.

For $0 < R < \theta r$ we claim

$$\frac{1}{|\partial B_R|} \int_{|h|=R} |u(P + h) - b_P(h_\parallel)|\, dS(h) \le CR/r. \tag{5.10}$$

To prove this, fix the first component h_ρ of $h_\parallel$ and write $\ell = h_t$. Each block radius with $h_\perp$ has the form $\sqrt{a_0^2 + |\xi|^2}$, where $a_0 \ge cr$ is the axial radius without $h_\perp$ and $|\xi| \le R$. The identity $\sqrt{a_0^2 + |\xi|^2} - a_0 = |\xi|^2/(\sqrt{a_0^2 + |\xi|^2} + a_0)$ bounds the radius error by CR^2/r. Since the map on the radii has bounded derivative, the transported radii have the same error. With h_ρ fixed, increasing ℓ in the radial plane decreases the transported first radius and increases the second, at rates bounded away from zero. This follows from $\|D\Phi - I\| < 1/4$ and the untransported rates $-1/\sqrt{2}$ and $1/\sqrt{2}$. Choose $A = C_1 R^2/r$ large enough to bracket the radius errors, then shrink θ_0 so that $C_1\theta_0 \le 1$. The planar points with parameters $\ell \pm A$ then stay in $B_{2\theta r}(P)$. Their transported radii bracket those of $\Phi(P + h)$ in the block order. Thus monotonicity bounds $u(P+h)$ between the planar values at $\ell - A$ and $\ell + A$, using actual planar values even when these points leave the disk defining the baseline. With h_ρ fixed, write $G(\ell)$ for the planar value, a nondecreasing function with $|G| \le 1$. For $R_1 = (R^2 - h_\rho^2)^{1/2}$, integrating $G(\ell+A) - G(\ell-A)$ over $[-R_1, R_1]$ gives the integral of G on $[R_1 - A, R_1 + A]$ minus its integral on $[-R_1 - A, -R_1 + A]$. The result is at most $4A$, including when G has jumps.

The sphere parametrization in Section 3 shows that, given h_ρ, the remaining coordinates are uniform on the sphere of radius R_1 in seven dimensions. The coordinate density there bounds the density of ℓ by C/R_1. The mean of $1/R_1$ is C/R, because the density of $z = h_\rho/R$ is proportional to $(1 - z^2)^{5/2}$ and $R_1^{-1} = R^{-1}(1 - z^2)^{-1/2}$. Thus the mean angular error is at most $CA/R \le CR/r$, proving (5.10). The same comparison holds for the

transported sign $w_F = 2\chi_F - 1$, since the cone sign has the same coordinate monotonicities.

After subtracting (5.9) from (5.4), the integrand can be written as

$$\big(u(P) - u(P+h)\big)\big(A_P(P+h) - A_{\Lambda_P}(h)\big) + \big(b_P(h_\parallel) - u(P+h)\big)A_{\Lambda_P}(h).$$

On $|h| < \theta r$ its integral has absolute value at most

$$Cm\theta^{1-\sigma} + C\frac{\delta|d|}{r}\int_{\mathbb{R}^8} |u(P) - u(P+h)|K(h)\,dh. \tag{5.11}$$

For the $|h|$ term in (5.8), a bounded increment gives $C(\delta/r)\int_0^{\theta r} a^{-\sigma}\,da$. The affine term gives the same integral by (5.10). Both are at most $Cm\theta^{1-\sigma}$. For the last term of (5.11), (5.5) gives $C(\delta|d|/r)(1+|d|)^{-\sigma}$, since the exact map has $D_P = |d|$. This is at most Cm for $|d| \le Cr$, and at most $C_T m r^{\sigma-1} = o(m)$ uniformly when $|d| \le T$. If $|t(P)| > \tau\rho(P)$, then

$$|t(\Phi(P))| \ge \tau r/\sqrt{2} - Ce_\lambda(r) \ge cr$$

by increasing L_0. Here $D_P \ge cr$, so (5.5) gives $O(r^{-\sigma})$. The near kernel is bounded by $C\delta K$, since (5.3) along the segment P to $P+h$ controls the difference of the kernel arguments by $C\delta|h|$ and the Jacobian error by $C\delta$. This again gives $O(m)$. The far bounds, including $\int_{|h|>\theta r} |A_{\Lambda_P}| \le C_\theta m$, control the compensated far integral in the exact region and the original far integral in its complement. Fixing one $\theta \in (0, \theta_0]$ now gives $|Lu - f(u)| \le Cm$ throughout the exterior, with a constant independent of r, λ.

Comparison with the curvature. We record a cutoff estimate used at two scales below. Suppose $G : \mathbb{R}^8 \to \mathbb{R}^8$ is a homeomorphism fixing zero with $\sup_{\overline{B}_R} |G(h) - h| \le \omega < R/2$. Then $B_{R-\omega} \subset G(B_R) \subset B_{R+\omega}$. The outer inclusion is immediate. For the inner inclusion, the segment from zero to a point in $B_{R-\omega}$ cannot leave $G(B_R)$ without meeting its boundary $G(\partial B_R)$. Every point of that boundary has norm at least $R - \omega$. Thus the symmetric difference of the image and B_R lies in a shell of volume $O(R^7\omega)$. After applying an invertible linear map with uniformly bounded norm and inverse norm, the shell remains at distances comparable to R and its volume is still $O(R^7\omega)$. Its integral against K is therefore $O(\omega R^{-1-\sigma})$.

Let $P = p + d\nu(p)$, $|d| \le T$, and $r = |P| \ge 2L_1\lambda$. Then $|p| \ge r - T \ge 3L_1\lambda/2$ for all sufficiently large λ. In particular $T \le c|p|$ eventually, so the tube construction applies and p, P have the same block directions. At p, $|t(p)| = \eta_\lambda(\rho(p)) \le \tau\rho(p)/4$ by the choice of L_0. Since t, ρ change by at most T from p to P, both points lie in the smaller region $|t| \le \tau\rho$ for large λ. The near estimates at p use the cutoff ratio $\theta r/|p| \le 2\theta$. By decreasing the allowed θ_0 here, this ratio meets the earlier restriction and $B_{2\theta r}(p)$ stays outside the core. The curvature of F exists at p by the cutoff estimate in

Section 3, since $F = E_\lambda$ near p. The sharp identity is

$$H_\sigma[F](p) = \mathrm{PV} \int_{\mathbb{R}^8} -w_F(Q) A_p(Q)\, dQ. \tag{5.12}$$

To justify (5.12), truncate both kernels by excluding $B_\varepsilon(p)$ in the Q variables. In the transformed term, change variables to $\Phi(Q)$. The sign becomes that of D, and the deleted set is $\Phi(B_\varepsilon(p))$. Center at $\Phi(p)$ and apply the cutoff estimate with

$$G(h) = D\Phi(p)^{-1}(\Phi(p+h) - \Phi(p)), \qquad \omega = O(\varepsilon^2)$$

by Taylor's formula. After applying $D\Phi(p)$, this replaces the deleted set by the centered differential ellipsoid at cost $O(\varepsilon^{1-\sigma})$. The difference between truncation by this ellipsoid and by a centered ball is supported in a ball of radius $C\varepsilon$. The tangent-plane sign integrates to zero for both truncations by central reflection. For the difference between the cone and tangent signs, (3.5) bounds the cost by $O(\varepsilon^{1-\sigma})$, since $\Phi(p) \in \Sigma \setminus \{0\}$. The cone integral with a centered spherical truncation is zero by Section 3. Letting $\varepsilon \downarrow 0$ gives (5.12).

Use the sharp planar baseline $b_p^\sharp(h_\parallel) = w_F(p + h_\parallel)$ for $|h_\parallel| < \theta r$, and zero outside. To compute its principal value integral against A_{Λ_p}, subtract the odd planar function $\mathrm{sgn}(\nu(p) \cdot h_\parallel)$. At small $h_\parallel$, $b_p^\sharp$ is the cap sign. Apply the signed-distance Taylor estimate of Section 3 at the displacement $h_\parallel$. The signs differ only where $|\nu(p) \cdot h_\parallel| \le C|h_\parallel|^2$. Since $\nu(p)$ belongs to the radial plane, $\nu(p) \cdot h_\parallel = \nu(p) \cdot h$, so the discrepancy lies in the slab $|\nu(p) \cdot h| \le C|h|^2$. The slab bound in (3.5) gives absolute integrability of the difference against K near zero. Elsewhere boundedness and the kernel tail suffice. The odd part cancels on spherical truncations, and the absolutely integrable part cancels by the perpendicular marginal. The baseline therefore has principal value zero, and subtracting it leaves the sharp near integrand

$$-w_F(p+h)\big(A_p(p+h) - A_{\Lambda_p}(h)\big) + \big(b_p^\sharp(h_\parallel) - w_F(p+h)\big) A_{\Lambda_p}(h).$$

The first term has integral bounded by $Cm\theta^{1-\sigma}$ by (5.8) at $d = 0$. The second has the same bound by (5.10) for w_F.

The diffuse far integral has three terms. Its integrand is

$$\begin{aligned} &u(P)\big(A_P(P+h) - A_{\Lambda_P}(h)\big) \\ &\quad - u(P+h) A_P(P+h) + b_P(h_\parallel) A_{\Lambda_P}(h). \end{aligned}$$

The sharp far integrand is

$$-w_F(p+h) A_p(p+h) + b_p^\sharp(h_\parallel) A_{\Lambda_p}(h).$$

To bound the coefficient of $u(P)$ in the diffuse far integral, observe that the $K(h)$ terms in $A_P(P+h) - A_{\Lambda_P}(h)$ cancel. Changing variables in the two remaining kernel terms writes the coefficient as the difference of integrals of K over the complements of the centered image of $B_{\theta r}(P)$ and the affine ellipsoid $\Lambda_P B_{\theta r}$. Apply the cutoff estimate with $R = \theta r$ and

$G(h) = \Lambda_P^{-1} O_P^{-1}(\Phi(P+h) - \Phi(P))$. The Taylor and radial-plane stretch estimates give $\omega = C\delta(R^2 + |d|R)/r$. For small θ_0 and large λ on the fixed normal band, $\omega < R/2$. The linear map $O_P\Lambda_P$ sends B_R to the affine ellipsoid, since the radial rotation leaves that ellipsoid invariant. The shell bound $C\omega R^{-1-\sigma}$ therefore gives the following bound for the coefficient

$$Cm\left(\theta^{1-\sigma} + \frac{|d|}{r}\theta^{-\sigma}\right). \tag{5.13}$$

For the far term with A_P, compare the two centers on the common integration region in the Q variables. For each fixed $\theta > 0$ and sufficiently large λ, every intermediate center on the segment from P to p has norm r' between $r - T$ and $r + T$, and is at distance at least $\theta r/2$ from that region because $|P - p| \le T$. This distance is at least $\theta r'/4$, so the integrated derivative estimate applies at this center with cutoff $\theta r'/4$. Since $m_\lambda(r')/r' \asymp m/r$, integrating along the segment gives an error $C_\theta m|d|/r$. The two truncations differ only in a shell about P of radius θr and thickness at most $2|d|$, of volume $O((\theta r)^7|d|)$. On this shell $|Q| \asymp r$ and both center distances are comparable to θr. The pointwise bound $C_\theta m r^{-8}$ gives the same cost there.

To replace u by w_F, the portions $|Q| < b_0 r$ and $|Q| > B_0 r$ cost at most $Cm(b_0^{7-\kappa} + B_0^{-\sigma})$. On the remaining annulus, (5.6) with $R = B_0 r$ bounds the L^1 error by $C_{B_0} r^{8-\sigma}$. Multiplication by the pointwise far kernel bound gives a cost $C_{\theta,b_0,B_0} m r^{-\sigma}$.

For the baseline term use the same radial plane and the same disk cutoff at p and P. On the common perpendicular subspace the dilations are the restrictions of $D\Phi$, so the second derivative bound in (5.3) along the segment gives $\|\Lambda_P - \Lambda_p\| \le C\delta|d|/r$. The determinant and kernel difference then cost $C\delta(|d|/r)K(h)$ pointwise by the mean value inequality. Integration on $|h| > \theta r$ gives $C_\theta m|d|/r$. Put $t_P(h_\parallel) = t(\Phi(P + h_\parallel))$ on the disk. Both block radii stay positive and the block directions stay fixed on this planar slice. Thus t has gradient e_t there, the unit vector for the second meridional coordinate. The chain rule and $\|D\Phi - I\| < 1/4$ give

$$\partial_{h_t} t_P = e_t \cdot D\Phi(P + h_\parallel) e_t \ge 3/4.$$

Moreover $|t_P - t_p| \le C|d|$ by the Lipschitz bound on $t \circ \Phi$. Consequently

$$|b_P - b_p^\sharp| \le C(1 + |t_P|)^{-\sigma} + 2\mathbf{1}_{\{|t_P| \le C|d|\}}$$

on the disk. For each fixed h_ρ, change variables from h_t to t_P. Since $\Phi(0) = 0$ and Φ is uniformly Lipschitz, the range of t_P lies in $[-Cr, Cr]$. The two terms then integrate to at most $Cr^{1-\sigma}$ and $C|d|$, respectively. The h_ρ interval has length $O(r)$, giving $\int |b_P - b_p^\sharp|\, dh_\parallel \le C(r^{2-\sigma} + |d|r)$. For $|h_\parallel| < \theta r$ one has

$$\int_{\{|h| > \theta r\}} K(h_\parallel + h_\perp)\, dh_\perp \le C(\theta r)^{-2-\sigma}.$$

If $|h_\parallel| \ge \theta r/2$, integrate over all six perpendicular variables and use the marginal (2.4) with $k = 2$. If $|h_\parallel| < \theta r/2$, the integration requires $|h_\perp| \ge$

$\sqrt{3}\theta r/2$, and the bound follows by radial integration in six dimensions. Since $|A_{\Lambda_p}| \le C\delta K$, the baseline replacement costs at most $C_\theta m(r^{-\sigma} + |d|/r)$.

Together these estimates give

$$\frac{|c_{8,s}^{-1}(Lu_\lambda - f(u_\lambda))(P) - H_\sigma[F_\lambda](p)|}{m} \le C\big(\theta^{1-\sigma} + b_0^{7-\kappa} + B_0^{-\sigma}\big) + C_{\theta,b_0,B_0,T}\big(r^{-\sigma} + r^{-1} + r^{\sigma-1}\big).$$

The first constant is independent of θ, b_0, B_0. Choose these cutoffs to make the first term arbitrarily small, then hold them and T fixed. The second term tends to zero uniformly on the band since $r \ge 2L_1\lambda$ and $0 < \sigma < 1$. This gives

$$(Lu_\lambda - f(u_\lambda))(P) = c_{8,s}H_\sigma[F_\lambda](p) + o(m_\lambda(r))$$

uniformly on each stated band.

Changing the interior data. Replacing u_λ by V_λ changes the residual at P by

$$c_{8,s}\int_{B_{L_1\lambda}} (u_\lambda - V_\lambda)(Q)K(Q-P)\,dQ.$$

The reaction is unchanged at P, and the kernel integral is well defined because $V_\lambda = u_\lambda$ near P and the core data are bounded and measurable. For Q in the core, $|Q-P| \ge r/2$. Thus the residual change is bounded by $C_{L_1}\lambda^8 r^{-8-\sigma}$, which is $O(m_\lambda(r))$ since $\lambda^8 r^{-8-\sigma}/m_\lambda(r) = (\lambda/r)^{7-\kappa}$. This proves the second assertion.

On the fixed normal band, p is outside the core for large λ, so F_λ and E_λ agree near p. In the exterior-minus-interior convention, their curvature difference $H_\sigma[E_\lambda](p) - H_\sigma[F_\lambda](p)$ is the ordinary integral of $(w_{F_\lambda} - w_{E_\lambda})(Q)K(Q-p)$ over the core. Thus, with $\mathcal{R}(w) = Lw - f(w)$, we have

$$\begin{aligned}&\mathcal{R}(V_\lambda)(P) - c_{8,s}H_\sigma[E_\lambda](p)\\ &\quad - \big(\mathcal{R}(u_\lambda)(P) - c_{8,s}H_\sigma[F_\lambda](p)\big)\\ &= c_{8,s}\int_{B_{L_1\lambda}} \Big[(u_\lambda - V_\lambda)(Q)K(Q-P) - (w_{F_\lambda} - w_{E_\lambda})(Q)K(Q-p)\Big]\,dQ.\end{aligned}$$

For Q in the core, its distance to any point on the segment from P to p is at least $r/2 - T \ge r/3$ for large λ. The mean value bound for K shows that replacing $K(Q-p)$ by $K(Q-P)$ costs at most $C_{L_1,T}\lambda^8 r^{-9-\sigma}$. Dividing by $m_\lambda(r)$ bounds this by $C_{L_1,T}(\lambda/r)^{7-\kappa}/r \to 0$ uniformly on the band.

With the same kernel, the difference of the two terms is $(u_\lambda - w_{F_\lambda}) - (V_\lambda - w_{E_\lambda})$. Its L^1 norm on the core is $o(\lambda^8)$. The first difference is $O_{L_1}(\lambda^{8-\sigma})$ by (5.6), and the second is $o(\lambda^8)$ by the core convergence in the third part of the proposition. Multiplying by the bound $Cr^{-8-\sigma}$ for the core kernel gives $o(m_\lambda(r))$, by the same ratio as above. This proves (5.1).

Finally, the stated curvature bound and (3.4) give

$$H_\sigma[E_\lambda](p) \le -c\lambda^{-\sigma}(1 + |p|/\lambda)^{-\kappa-1-\sigma} \le -c' m_\lambda(r).$$

Indeed $r \geq 2L_1\lambda$ and $|p| \leq r + T \leq 2r$ for large λ, so $c' > 0$ can be chosen independently of T. Absorbing the uniform error for sufficiently large λ gives the strict residual bound. □

6. The diffuse minimizer in eight dimensions

Proposition 7. *For each $0 < s < 1/2$, the cap (3.1) yields a smooth, sign-changing entire minimizer $v : \mathbb{R}^8 \to (-1, 1)$ of (2.1). It solves (1.1), is invariant under $O(4) \times O(4)$, and satisfies (1.2).*

Write the saddle from Section 4 as $S(\rho, t)$. We use its block monotonicity, its strict increase in t on $(-\rho, \rho)$, and its radial monotonicity $S_\rho \geq 0$ for $t > 0$. Lemma 5 supplies the estimate

$$|S_\rho(\rho, t)| \leq C/\rho \qquad (|t| < \rho). \tag{6.1}$$

6.1. **Gluing the profiles.** Let d_λ denote the signed distance to ∂E_λ, positive in E_λ, and set $\mathcal{R}(V) = LV - f(V)$. Dilation gives $d_\lambda(P) = \lambda d_1(P/\lambda)$. The signed distance is 1-Lipschitz. For two points on the same side this follows from the ordinary distance bound, and for points on opposite sides the segment between them meets the boundary, so the sum of their distances is at most the length of the segment.

Apply Proposition 6 with $v = S$ and $E_1 = E$. The boundedness, block monotonicity, and uniform end limits of S are proved in Section 4. The smooth cap E is contained in D and is invariant under the block rotations by its defining inequality (3.1). Since $\rho \geq |P|/\sqrt{2}$, the description $t > \eta(\rho)$ from Section 3 applies throughout the exterior of a sufficiently large ball. The expansion (3.2) gives $\eta(\rho) \asymp \rho^{-\kappa}$ and $|\eta^{(j)}(\rho)| \leq C_j \rho^{-\kappa-j}$ for $1 \leq j \leq 3$, with $\kappa = 5/2 < 7$. Take L_0 to be an exterior radius for its map Φ_λ. On the cap with $|p| \geq L_0\lambda$ the arclength label is comparable to $\rho(p)$ by Proposition 6, since $\rho(p) \geq |p|/\sqrt{2}$. The height derivative $\eta'_\lambda(\rho(p))$ is negative off the axis by the formula for η' in Section 3. Then fix $L_1 > 4L_0$. The map is the same for all $L_1 \geq L_0$, so its exact tube and the equality $\Phi_\lambda^{-1}(D) = E_\lambda$ apply beyond $L_0\lambda$, including in the blending annulus below.

On $\partial E \cap \overline{B}_{4L_1}$, compactness and the tubular-neighborhood theorem for the smooth embedded boundary give a common tubular radius $4\delta > 0$. One may construct the tube on a slightly larger portion of the boundary. The positive gap between the two radii excludes any closest point outside that portion when δ is small. After dilation, points in the tube have a unique smooth footpoint $p = p(P)$ and $P = p + d_\lambda(P)\nu(p)$, with ν the inward normal. The footpoint scales in the same way as the signed distance. By shrinking the unit tube to lie within the region of smoothness, its derivatives are bounded on the compact portion in use. Hence $D^2 d_\lambda = O(\lambda^{-1})$ and the first derivative of $p(P)$ is uniformly bounded. By the separation in Section 3, we can take the tube inside D. We also require $4\delta < 1$, using $\operatorname{dist}(0, \overline{E}) = 1$ from that section.

On this compact portion of the dilated cap, define a smooth radial label $r_\lambda(p)$. Take it equal to $c_1\lambda$ for $|p| \le L_0\lambda$ and to the arclength label $S_\lambda(\rho(p))$ of Proposition 6 for $|p| \ge 2L_0\lambda$, using a smooth convex interpolation in $|p|/\lambda$. The label is constant near the factor axis, and wherever the arclength term is used both block radii are positive. It is thus smooth and satisfies $r_\lambda(p) = \lambda r_1(p/\lambda)$. The unit-scale label is positive on the compact portion, so the scaled label is comparable to λ and has uniformly bounded first derivative along the cap.

Where the arclength label has positive weight, $|p| \ge L_0\lambda$. By taking L_0 large we have $\eta_\lambda(\rho(p)) \le \rho(p)/2$. Both block radii at p are therefore at least $\rho(p)/(2\sqrt{2}) \ge L_0\lambda/4$. Reduce δ so that $4\delta < L_0/8$. Each radius is 1-Lipschitz, so neither can vanish along the normal segment of length at most $4\delta\lambda$. The block directions stay fixed on this segment. The inward normal's radial component is $-\eta_\lambda'/\sqrt{1+(\eta_\lambda')^2} \ge 0$, so $\rho(P) \ge \rho(p)$ for a positive normal displacement. Since $S_\lambda(\rho(p)) \le \rho(p)$, the arclength label is at most $\rho(P)$. The constant label has the same property for sufficiently small $c_1 > 0$. In fact $\rho(p) \ge \lambda/\sqrt{2}$ on the cap and ρ changes by at most $4\delta\lambda$ in the tube. We can take $4\delta < 1/(2\sqrt{2})$ and $c_1 < 1/(2\sqrt{2})$. Thus $r_\lambda(p) \le \rho(P)$ when $d_\lambda(P) \ge 0$. After fixing the labels, reduce δ so that $4\delta\lambda < r_\lambda(p)$ throughout the compact tube and $\delta < L_1/2 - 2L_0$. Also require $\delta < 2cL_0$ for the tube constant c in Proposition 6.

For $|P| \le 3L_1\lambda$, define I_λ by blending $S(r_\lambda(p), d_\lambda(P))$ with the phase background. Give the saddle profile a smooth weight in $[0,1]$ depending on d_λ/λ, equal to one for $|d_\lambda| \le \delta\lambda$ and zero for $|d_\lambda| \ge 2\delta\lambda$. Use only the background where the weight is zero. The background is -1 for $d_\lambda < 0$ and $1 - C_1\lambda^{-\sigma}$ for $d_\lambda > 0$. Near the cap the formula uses only the smooth normal profile. Where this profile is used, the signed distance and footpoint are smooth. Outside this tube the formula is locally constant on each phase. The smooth cutoff joins these formulas within the tube. When a footpoint is needed, its radius is at most $3L_1\lambda + 2\delta\lambda < 4L_1\lambda$, and the profile argument satisfies $|d_\lambda| < r_\lambda(p)$.

In the positive phase of this ball, the separation from the cone in Section 3 gives $t(P) \ge c\lambda$. The well estimate then gives $1 - S(P) \le C\lambda^{-\sigma}$. Choose $C_1 > C$, and take λ sufficiently large that the positive background belongs to $(-1, 1)$. Both backgrounds are strictly below $S(P)$ where used.

The normal profile is strictly below $S(P)$. For nonpositive normals the profile is nonpositive, while $S(P) > 0$ since the compact tube lies in D. On a positive normal, $P \in E_\lambda$. A nearest cone point is outside $\overline{E}_\lambda$ by the separation in Section 3. The segment to this point crosses ∂E_λ before the endpoint, so $d = d_\lambda(P) < t(P)$. With the bound $r_\lambda(p) \le \rho(P)$ and $S_\rho \ge 0$, we have

$$S(r_\lambda(p), d) \le S(\rho(P), d) < S(\rho(P), t(P)).$$

The first comparison integrates $S_\rho \ge 0$ over radii at the fixed positive coordinate d, which is smaller than the starting label $r_\lambda(p)$. For the strict

inequality, choose an intermediate normal coordinate between d and $t(P)$ inside $(-\rho(P), \rho(P))$ and use strict increase up to that coordinate. This also covers the case $t(P) = \rho(P)$ on an axis.

Take a smooth radial cutoff $\xi_\lambda(P)$ equal to one for $|P| \le L_1\lambda/2$ and zero for $|P| \ge L_1\lambda$, with values in $[0,1]$ and depending on $|P|/\lambda$. Define V_λ as the blend of I_λ with $S \circ \Phi_\lambda$, using ξ_λ for the inner weight. Beyond its support use the transported saddle alone. Then V_λ is locally Lipschitz and $-1 \le V_\lambda \le S$, by the inner comparison and Proposition 6. It is jointly continuous in (λ, P), since the geometry maps and cutoffs are scaled from fixed maps and the background depends continuously on λ.

For $|P| \ge L_1\lambda/2$ in the compact tube and $|d_\lambda| \le \delta\lambda$, the footpoint satisfies $|p| \ge L_1\lambda/2 - \delta\lambda > 2L_0\lambda$. The choice of δ ensures $|d_\lambda| \le c|p|$. Thus the exact transport uses the same label and normal distance as the inner profile. In particular,

$$V_\lambda(P) = S(r_\lambda(p), d_\lambda(P)) \qquad (|P| \le 3L_1\lambda,\ |d_\lambda(P)| \le \delta\lambda). \tag{6.2}$$

This identity also covers the pure transport part of the compact tube beyond the blending annulus. Where the exterior transport is exact, it also strictly decreases the saddle value. Along the segment joining the two pairs of block radii, both radii stay positive, the first strictly increases, and the second strictly decreases. The segment meets the cone at most once because the difference of the radii strictly decreases. In either open chamber, Section 4 gives $S_a < 0 < S_b$ when both radii are positive. The derivative of S along the segment is therefore negative off the cone, giving the strict decrease by integration.

In the blend and the exterior,

$$|t(\Phi_\lambda(P))| \ge c\,|d_\lambda(P)| \qquad (|P| \ge L_1\lambda/2). \tag{6.3}$$

Indeed Φ_λ takes $\partial(\Phi_\lambda^{-1}D)$ to Σ, so

$$|t(\Phi_\lambda(P))| = \inf_{Q \in \partial(\Phi_\lambda^{-1}D)} |\Phi_\lambda(P) - \Phi_\lambda(Q)| \ge c \inf_{Q \in \partial(\Phi_\lambda^{-1}D)} |P - Q|$$

by the Lipschitz bound for the inverse. A nearest point exists because the preimage boundary is nonempty and closed. A nearest point in $\{|Q| > L_0\lambda\}$ lies on ∂E_λ, since the two sets agree on that open exterior region. If the nearest point lies in $\{|Q| \le L_0\lambda\}$, its distance to P is at least $|P| - L_0\lambda \ge |P|/2$. This also bounds the cap distance up to a constant, since for a fixed $p_* \in \partial E$, $|d_\lambda(P)| \le |P - \lambda p_*| \le C|P|$. The signs of $t(\Phi_\lambda(P))$ and $d_\lambda(P)$ agree because $\Phi_\lambda^{-1}(D)$ and E_λ agree in the exterior.

For $\varepsilon \in \{-1, 1\}$ and $\varepsilon d_\lambda(P) > 0$, the well estimate, the background values and (6.3) give the single bound

$$|V_\lambda(P) - \varepsilon| \le C(1 + |d_\lambda(P)|)^{-\sigma} + C\lambda^{-\sigma}. \tag{6.4}$$

For the inner profile, this is the saddle estimate at signed cone distance $d_\lambda(P)$, and on the blend it follows by convexity of the bound. In particular $V_\lambda(\lambda X) \to 2\mathbf{1}_E(X) - 1$ off ∂E. This holds also for a sequence of scaled points

converging off ∂E, by the scaling and continuity of the signed distance. The boundary has measure zero, so bounded convergence after scaling gives the core L^1 hypothesis of Proposition 6.

6.2. **Residual on bounded scaled regions.** Suppose $|P| \le 2L_1\lambda$ and $|d_\lambda(P)| \le \delta\lambda/4$, and write $P = p + d\nu(p)$. Put $r_0 = r_\lambda(p)$ and define $F(z) = S(r_0, z)$, extending it by the endpoint values for $|z| \ge r_0$. This is a nondecreasing Lipschitz function with values in $[-1, 1]$, by the monotonicity and global gradient bound for S. Choose a cone normal comparison point X_0 with radial coordinates (r_0, d) and both block directions equal to a fixed unit vector in $\mathbb{R}^4$. Write ν_0 for the inward cone normal in these directions. By (6.2), $V_\lambda(P) = S(X_0) = F(d)$. In the residual, we can therefore replace $f(V_\lambda(P))$ by $LS(X_0)$. The two operator integrals have the same center value $F(d)$. The planar profiles $F(d + \nu(p)\cdot h)$ and $F(d + \nu_0 \cdot h)$ give equal operator values at $h = 0$, since the kernel is rotation invariant. Subtracting these equal planar values from the two operator integrals yields

$$
\begin{aligned}
c_{8,s}^{-1}\mathcal{R}(V_\lambda)(P) &= \int_{\mathbb{R}^8} [D_\lambda(h) - D_\lambda^0(h)]|h|^{-8-\sigma}\, dh, \\
D_\lambda(h) &= F(d + \nu(p)\cdot h) - V_\lambda(P + h), \\
D_\lambda^0(h) &= F(d + \nu_0 \cdot h) - S(X_0 + h).
\end{aligned} \tag{6.5}
$$

The profiles are locally Lipschitz at the evaluation points and the integrals converge absolutely since $\sigma < 1$.

Choose $b_0 > 0$ small enough that $b_0 \le \min(\delta/4, L_1/2)$. Then, for $|h| = a \le b_0\lambda$, the segment from P to $P + h$ remains in the region of (6.2), by the distance and radius bounds. The derivative of the footpoint and label maps gives a label change of at most Ca. The same estimate holds for the cone comparison since ρ is 1-Lipschitz. For b_0 sufficiently small, the cone comparison stays away from both axes. In both comparisons, the initial label satisfies $r_0 > 4\delta\lambda$, the label changes by at most Ca, and the absolute normal coordinate is at most $|d| + a \le \delta\lambda/2$. By reducing b_0 so that $Cb_0 < \delta$, every intermediate label remains larger than the absolute normal coordinate and comparable to λ. Holding the normal coordinate fixed and replacing the variable label by r_0 thus costs $O(a/\lambda)$, by (6.1). In the cap tube $\nabla d_\lambda(P) = \nu(p)$ and $\|D^2 d_\lambda\| \le C/\lambda$. The cone distance has gradient ν_0 at X_0 and the same Hessian bound nearby: the two radii of X_0 are $(r_0 - d)/\sqrt{2}, (r_0 + d)/\sqrt{2}$, each comparable to λ, and each block norm has Hessian bounded by the reciprocal of its radius. Thus the distance differs from its planar argument by $O(a^2/\lambda)$.

Let ν be the unit normal of the comparison in use and put $z = d + \nu \cdot h$, $\omega = Ca^2/\lambda$. By monotonicity, the error from replacing the actual distance by z is bounded by $F(z + \omega) - F(z - \omega)$. On $|h| = a$, the coordinate density in Section 3 bounds the density of z by C/a on $[d - a, d + a]$. The one-dimensional shift estimate used in (5.10) bounds the integral of $F(z + \omega) -$

$F(z-\omega)$ on this interval by 4ω. Thus the angular average of the normal-coordinate error is at most $C\omega/a \le Ca/\lambda$, the same order as the radial label error. It follows that the contribution from $|h| < b\lambda$ in (6.5) for $0 < b \le b_0$ is at most

$$C\int_0^{b\lambda} (a/\lambda)a^{-1-\sigma}\,da = C'b^{1-\sigma}\lambda^{-\sigma}.$$

The outer integral is bounded by $C(b\lambda)^{-\sigma}$ by boundedness. Taking $b = b_0$ gives the absolute residual estimate.

Now fix M and restrict to $|d| \le M$, which lies in the comparison tube $|d| \le \delta\lambda/4$ for all large λ. Along any sequence of these evaluation points with $\lambda \to \infty$, compactness gives a subsequence with $p/\lambda \to p_\infty$. By continuity and scaling of the inward normal and the label, $\nu(p) \to \nu_\infty = \nu_E(p_\infty)$ and $r_0/\lambda \to r_\infty = r_1(p_\infty) > 0$. It follows that $P/\lambda \to p_\infty$. The coordinates of X_0/λ tend to $(r_\infty, 0)$ with the fixed block directions, so $X_0/\lambda \to X_\infty \in \Sigma \setminus \{0\}$. After putting $h = \lambda k$, the values $V_\lambda(P + \lambda k)$ converge to the cap sign at $p_\infty + k$ away from ∂E by (6.4). Similarly $S(X_0 + \lambda k)$ converges to the cone sign at $X_\infty + k$ off Σ. When the limiting planar normal component of k is nonzero, the argument of F has that sign and magnitude at least $c(k)\lambda$. Clamping the argument to $[-r_0, r_0]$ preserves the sign and a lower bound of the same order, since $r_0/\lambda \to r_\infty > 0$. The clamped argument is exactly the cone distance in the saddle profile defining F. The well estimate of Lemma 4 therefore gives convergence of the planar profiles to their half-space signs. Write $w_G = 2\mathbf{1}_G - 1$. Outside the exceptional interfaces and planes, which have measure zero, $D_\lambda(\lambda k) - D^0_\lambda(\lambda k)$ in (6.5) converges to

$$\operatorname{sgn}(\nu_\infty \cdot k) - w_E(p_\infty + k) - \operatorname{sgn}(\nu_0 \cdot k) + w_D(X_\infty + k).$$

The kernel and volume contribute the factor $\lambda^{-\sigma}$ under $h = \lambda k$. On $|k| \ge b$, dominated convergence applies to the resulting integral with majorant $C|k|^{-8-\sigma}$. Both planar signs integrate to zero by reflection, and the cone sign integrates to zero by the centered truncation identity in Section 3. The remaining term $-w_E(p_\infty + k)$ gives the exterior-minus-interior curvature truncated at b.

For fixed $b > 0$, the truncated cap curvature depends continuously on the footpoint by dominated convergence in variables centered there, since ∂E has measure zero. Estimate (3.5) gives a truncation error $O(b^{1-\sigma})$, uniformly over the compact set of footpoints by the Hessian bound for the unit signed distance. Thus the curvature is continuous there as a uniform limit of the truncated integrals. The deleted diffuse terms have the same bound after multiplication by λ^σ. Letting $b \downarrow 0$ yields the uniform limit

$$\lambda^\sigma \mathcal{R}(V_\lambda)(P) = c_{8,s} H_\sigma[E](p/\lambda) + o(1) \qquad (|d| \le M,\ |P| \le 2L_1\lambda). \tag{6.6}$$

Indeed, on any sequence violating uniform convergence, the subsequence above gives convergence of the scaled residual to $c_{8,s}H_\sigma[E](p_\infty)$. Continuity gives the same limit for the curvature at the moving footpoint p/λ, contradicting the violation.

For $|P| \le 2L_1\lambda$ and $|d_\lambda(P)| \ge \delta\lambda/4$, a ball of radius $c\lambda$ about P, with $c \le \min(\delta/8, L_1/2)$, lies in the same phase and has cap distance at least $\delta\lambda/8$, by the Lipschitz bound for signed distance. It also stays within $|Q| < 3L_1\lambda$, where the compact profile is defined. The well error is $O(\lambda^{-\sigma})$ there by (6.4). We claim the Lipschitz bound on that ball is also $O(\lambda^{-\sigma})$. For the inner profile, its ρ, t derivatives are directional derivatives of S in an orthonormal radial plane. They have this bound by Lemma 4, since the distance argument is at least $\delta\lambda/8$ in absolute value. The label and normal coordinate have bounded derivatives where this profile is used. For $S \circ \Phi_\lambda$, use (6.3), the same well estimate, and the uniform bound on $D\Phi_\lambda$. The backgrounds are constant in the ball's phase. In each convex combination, derivatives of the cutoffs are $O(\lambda^{-1})$ and multiply differences between two values within $O(\lambda^{-\sigma})$ of the same well, so they preserve the derivative bound. Integrating along each segment in the ball gives the claimed Lipschitz bound there. Split the integral for $LV_\lambda(P)$ as in (5.5), at radii 1 and $c\lambda$. The Lipschitz bound, the well error, and the bounded increments in the tail give $O(\lambda^{-\sigma})$. The reaction is $O(\lambda^{-\sigma})$ as f vanishes at that well. This gives the same absolute residual bound throughout the rest of the compact scaled region.

6.3. **Correction in the pure phases.** For the cap of Section 3, $\beta = \kappa + 1 + \sigma = 7/2 + \sigma < 8$. Put

$$\widetilde{m}_\lambda(P) = \lambda^{\beta-\sigma}(\lambda^2 + |P|^2)^{-\beta/2}.$$

On $|P| \le 2L_1\lambda$ this weight is comparable to $\lambda^{-\sigma}$. On $|P| \ge 2L_1\lambda$ it is comparable to the exterior weight $\lambda^{\kappa+1}|P|^{-\kappa-1-\sigma}$ of Proposition 6, since $\beta - \sigma = \kappa + 1$ and $(\lambda^2 + |P|^2)^{1/2} \asymp |P|$ there.

The compact absolute residual bound and the second assertion of Proposition 6 give the global absolute bound below. For the sign on a fixed band $|d_\lambda| \le M$, first consider $|P| \le 2L_1\lambda$. Its footpoint is in $|p| \le 3L_1\lambda$ for large λ, and (3.3) bounds the curvature at p/λ above by a strictly negative constant on this compact set. Thus (6.6) gives a negative bound of order $\lambda^{-\sigma}$ with a coefficient independent of M. For $|P| \ge 2L_1\lambda$ on the same band, choose a nearest point $p \in \partial E_\lambda$, which exists by closedness. Minimizing squared distance along the smooth boundary makes $P - p$ normal at p. If $P \ne p$, the open segment from p to P stays on one side, since any intervening boundary point would be closer to P. The normal sign thus agrees with $d_\lambda(P)$, so $P = p + d_\lambda(P)\nu(p)$. Since $|p| \ge |P| - M$, for large λ we have $|d_\lambda(P)| \le c|p|$, as required by the exact tube in Proposition 6. Its signed residual bound applies by (3.3) and the equivalent bound in $|p|$ from Section 3. Combining the two regions gives

(6.7)
$$|\mathcal{R}(V_\lambda)| \le C_{\mathrm{res}}\widetilde{m}_\lambda \quad \text{everywhere}, \qquad \mathcal{R}(V_\lambda) \le -c_{\mathrm{res}}\widetilde{m}_\lambda \quad \text{on } |d_\lambda| \le M.$$

The absolute bound holds for all sufficiently large λ. The constants are independent of M, and the lower threshold on λ for the signed estimate

may depend on M. Fix $a_0 \in (1/\sqrt{3}, 1)$. By (6.4) we can choose T_0 and then a lower bound on λ such that $|V_\lambda| \geq a_0$ on $|d_\lambda| \geq T_0$.

Take a smooth even $0 \leq \chi \leq 1$ with $\chi = 0$ on $[-1, 1]$ and $\chi = 1$ outside $(-2, 2)$. For $0 < T \leq \lambda$ one has

$$\left|L(\widetilde{m}_\lambda \chi(d_\lambda/T))(P)\right| \leq CT^{-\sigma}\widetilde{m}_\lambda(P). \tag{6.8}$$

Write $R_P = (\lambda^2 + |P|^2)^{1/2}$. The signed-distance Lipschitz bound controls the cutoff increment by $C\min(1, |P-Q|/T)$. For $|P-Q| \leq R_P/4$, the quantity R_Z is comparable to R_P on the segment from P to Q, and $|\nabla \widetilde{m}_\lambda(Z)| \leq C\widetilde{m}_\lambda(P)/R_P$ there. The product increment is therefore bounded by

$$C\widetilde{m}_\lambda(P)\left(\frac{|P-Q|}{R_P} + \min(1, |P-Q|/T)\right).$$

Integrating the first term against the radial kernel gives $CR_P^{-\sigma}\widetilde{m}_\lambda(P)$. For the second, the radial integral

$$\int_0^\infty \min(1, r/T)\, r^{-1-\sigma}\, dr = \left(\frac{1}{1-\sigma} + \frac{1}{\sigma}\right) T^{-\sigma}$$

gives the required cutoff bound.

At distances $|P-Q| \geq R_P/4$, bound the increment by $\widetilde{m}_\lambda(P) + \widetilde{m}_\lambda(Q)$. The first weight contributes $CR_P^{-\sigma}\widetilde{m}_\lambda(P)$ by the kernel tail. For the second weight on $|Q| \leq 2R_P$, bound the kernel by $CR_P^{-8-\sigma}$ and use

$$\int_{|Q|\leq 2R_P} \widetilde{m}_\lambda(Q)\, dQ \leq C\lambda^{\beta-\sigma} R_P^{8-\beta},$$

which follows by integrating $\lambda^{\beta-\sigma}|Q|^{-\beta}$ and using $\beta < 8$. On $|Q| > 2R_P$, use $|P-Q| \geq |Q|/2$ to bound the integral by $C\lambda^{\beta-\sigma}\int_{2R_P}^\infty r^{-1-\beta-\sigma}\, dr$. Both estimates give $CR_P^{-\sigma}\widetilde{m}_\lambda(P)$. Since $R_P \geq \lambda \geq T$, the near and far bounds prove (6.8).

Choose $1/\sqrt{3} < a_1 < a_0$ and $c_w = 3a_1^2 - 1 > 0$. First choose A so $c_w A > 4C_{\mathrm{res}}$. Then choose $T \geq T_0$ so that $C_{\mathrm{cut}} A T^{-\sigma} < \min(c_{\mathrm{res}}, c_w A)/2$, with C_{cut} from (6.8). These choices are possible because the residual constants are independent of the band width. With A, T fixed, take λ large enough for $\lambda \geq T$ and the signed estimate on $|d_\lambda| \leq 2T$. Also take λ large enough that the shift $\zeta = A\widetilde{m}_\lambda \chi(d_\lambda/T)$ is at most $a_0 - a_1$, since $\widetilde{m}_\lambda \leq \lambda^{-\sigma}$. If $\zeta(P) > 0$, then $|d_\lambda(P)| > T \geq T_0$. Each value on the segment from $V_\lambda(P) - \zeta(P)$ to $V_\lambda(P)$ has absolute value at least $|V_\lambda(P)| - \zeta(P) \geq a_1$. Hence $f'(a) \leq -c_w$ on that segment for the cubic on $\mathbb{R}$. The mean value bound $f(V_\lambda) - f(V_\lambda - \zeta) \leq -c_w\zeta$ follows and also holds where $\zeta = 0$. Consequently

$$\mathcal{R}(V_\lambda - \zeta) \leq \mathcal{R}(V_\lambda) + C_{\mathrm{cut}} A T^{-\sigma}\widetilde{m}_\lambda - c_w\zeta.$$

For $|d_\lambda| \leq 2T$, use (6.7) with $M = 2T$ to bound this by $-c_{\mathrm{res}}\widetilde{m}_\lambda/2$. For $|d_\lambda| > 2T$, we have $\zeta = A\widetilde{m}_\lambda$ and the bound is at most $(C_{\mathrm{res}} - c_w A/2)\widetilde{m}_\lambda < 0$.

Thus $\mathcal{R}(V_\lambda - \zeta) \le -\gamma\widetilde{m}_\lambda$ everywhere, with $\gamma = \min\{c_{\mathrm{res}}/2, c_w A/2 - C_{\mathrm{res}}\} > 0$. Set

$$a_\lambda = \max\{-1, V_\lambda - \zeta\}. \tag{6.9}$$

The clipped function is jointly continuous in (λ, P) by the scaling of d_λ, the formula for ζ, and joint continuity of V_λ. For each λ it is locally Lipschitz, since V_λ and ζ are, and $-1 \le a_\lambda \le S < 1$. Its kernel integral converges absolutely by the local Lipschitz bound, boundedness, and $\sigma < 1$. At a point where $a_\lambda > -1$, the center value and reaction are unchanged by clipping. The increments can only decrease because the values at the other points increase. Where $a_\lambda = -1$, every increment is nonpositive and the reaction vanishes. Hence a_λ is a pointwise subsolution, with $\mathcal{R}(a_\lambda) \le -\gamma\widetilde{m}_\lambda$ where $a_\lambda > -1$.

6.4. **Sliding and exhaustion.** For all sufficiently large λ one has $a_\lambda < S$ everywhere. If the value is clipped it equals $-1 < S$, and where the subtraction is positive and the value is unclipped the order follows from $V_\lambda \le S$. If $\zeta = 0$, then $|d_\lambda| \le 2T$. Take λ so large that $2T \le \delta\lambda$. In the compact region $|P| \le 2L_1\lambda$, (6.2) gives the strictly ordered normal profile. Beyond that region the nearest cap point gives the exact exterior tube representation, as in the proof of (6.7). The strict order of the transported saddle then gives $a_\lambda < S$ there as well.

For P in a fixed compact set, $P/\lambda \to 0$ uniformly. Since $d_1(0) = -1$ and $4\delta < 1$, the distance scaling gives $d_\lambda(P) < -2\delta\lambda$ there for large λ. Also $|P| < L_1\lambda/2$, so V_λ equals its negative background and clipping gives $a_\lambda = -1$.

For a positive witness, fix a sufficiently large λ_0 and take $p_0 = (0, \lambda_0 e_1)$ on the cap axis, with e_1 a unit vector in the y-block. The inward normal is $(0, e_1)$ by Section 3. Choose $0 < d_* < \min\{T, \delta\lambda_0\}$ so small that $P_0 = p_0 + d_*(0, e_1)$ lies in $B_{L_1\lambda_0/2}$. In the compact tube its footpoint is p_0 and $d_{\lambda_0}(P_0) = d_*$, so $V_{\lambda_0}(P_0) = S(r_{\lambda_0}(p_0), d_*) > 0$. The correction vanishes there, hence $a_{\lambda_0}(P_0) > 0$.

Minimize with exterior datum S on B_R^8, and denote the result by v_R. Lemma 2 applies. For the slide we also need continuity up to the boundary. Set $w = v_R - S$, which vanishes almost everywhere outside B_R. Its interior H^s seminorm is finite by the minimizer energy and the smoothness of S. Its cross seminorm is finite by boundedness and the finite cross interaction from Lemma 2. Since w is bounded with bounded support, $w \in H^s(\mathbb{R}^8)$.

By the distributional equations for v_R and S, $Lw = g$ in B_R, where $g = f(v_R) - f(S)$ is bounded. For $\varphi \in C_c^\infty(B_R)$, the Fourier symbol of L and Plancherel give the identity

$$\int_{\mathbb{R}^8} (-\Delta)^{s/2} w \, (-\Delta)^{s/2}\varphi = \langle Lw, \varphi\rangle = \int_{B_R} g\varphi.$$

Both sides are continuous in the $H^s(\mathbb{R}^8)$ norm of the test. For an H^s test φ vanishing almost everywhere outside B_R, the compression $\varphi(P/(1-\epsilon))$ has

support in $\overline{B}_{(1-\epsilon)R}$ and tends to φ in H^s. This follows first for smooth compactly supported functions, then by density and the uniform norm bounds for dilations near the identity. In fact, dilation by r scales the L^2 norm by r^4 and the H^s seminorm by r^{4-s}. Mollify the compressed test with radius smaller than $\epsilon R/2$ and small enough that its H^s error is at most ϵ. The resulting smooth tests are supported in B_R, so the identity extends to all zero-exterior tests as in [6, Definition 2.1].

The ball is a bounded Lipschitz domain satisfying the exterior ball condition. Thus [6, Proposition 1.1] gives a $C^s(\mathbb{R}^8)$ representative of w. Since $w = 0$ almost everywhere on B_R^c, this representative vanishes throughout B_R^c, including the boundary. We use the corresponding continuous representative $v_R = w + S$. For a solution valued in $[-1, 1]$, contact with a well at an interior point forces the solution to equal that well almost everywhere. Indeed, $v_R + 1$ at a contact with -1, or $1 - v_R$ at a contact with 1, satisfies the contact assertion of Lemma 3 since $f(\pm 1) = 0$. The exterior datum $S \in (-1, 1)$ excludes either constant. In particular $v_R > -1$ on $\overline{B}_R$.

For fixed R, define $\mathcal{M}_R(\lambda) = \max_{\overline{B}_R}(a_\lambda - v_R)$. On each compact scale interval, joint continuity of a_λ gives uniform continuity on the product with $\overline{B}_R$, and hence continuity of $\mathcal{M}_R$. It is negative for sufficiently large λ, since then $a_\lambda = -1$ on $\overline{B}_R$ and $v_R > -1$ there. If $\mathcal{M}_R(\lambda_0) \geq 0$, there is a scale with $\mathcal{M}_R(\lambda) = 0$. Boundary continuity and $a_\lambda < S$ put the contact point in B_R. The global difference has a maximum of zero there, because $a_\lambda < S = v_R$ on the exterior. Thus $L(a_\lambda - v_R) \geq 0$ at contact. The contact value is $v_R > -1$, so the strict subsolution inequality and equality of the reactions give $L(a_\lambda - v_R) < 0$, a contradiction. This proves $a_{\lambda_0} \leq v_R$. Let $\iota(x, y) = (y, x)$. The transformed function $P \mapsto -v_R(\iota P)$ is a minimizer with exterior datum S, because the energy is invariant under block swap and negation and $-S(\iota P) = S(P)$. Applying the lower bound to it gives

$$a_{\lambda_0}(P) \leq v_R(P) \leq -a_{\lambda_0}(\iota P).$$

For any $G \in O(4)\times O(4)$, the pullback $v_R \circ G$ minimizes on B_R with the same datum, since G preserves the ball, the kernel, and S. The two minimizers are ordered almost everywhere by Lemma 3. The change of variables $Q = GP$ gives $\int_{B_R} v_R \circ G = \int_{B_R} v_R$. The ordered functions must therefore agree almost everywhere on B_R. They agree pointwise there by interior continuity, and they have the same exterior datum.

Take $R \to \infty$ along the integers. On each fixed closed ball, Lemma 2 gives bounds through any order $j + 1$ on a slightly larger ball, uniformly for all sufficiently large R. The bound on derivatives of order $j+1$ makes the derivatives of order j equicontinuous by integration along segments. Arzelà–Ascoli and a diagonal subsequence over the balls and derivative orders give smooth convergence on compact sets to v. The derivative limits agree with those of v by the fundamental theorem of calculus on segments. For any bounded Lipschitz open set A, eventually $A \subset B_R$. Restriction gives minimality of v_R on A, and the closure assertion of the same lemma gives minimality of

v on A. Thus v is an entire minimizer. The first variation and interior regularity in Lemma 2, applied on each ball, give $Lv = f(v)$ pointwise. The two bounds give $v(P_0) > 0 > v(\iota P_0)$. The same well-contact argument used for v_R excludes $v = \pm 1$ at any point, since v is sign-changing. The block symmetries pass to the limit, as do both barrier bounds.

Fix the scale λ_0 of those bounds. Outside $B_{L_1\lambda_0}$ the uncorrected profile is $S \circ \Phi_{\lambda_0}$. By (5.3) and the global Lipschitz bound on S,

$$|S(\Phi_{\lambda_0}(P)) - S(P)| \le C_{\lambda_0}(1 + |P|)^{-\kappa}.$$

The correction satisfies $0 \le \zeta(P) \le C_{\lambda_0}(1+|P|)^{-\beta}$, where $\beta = \kappa+1+\sigma > \kappa$. Clipping at -1 is 1-Lipschitz and fixes $S(P)$. Thus $|a_{\lambda_0}(P) - S(P)| \le C_{\lambda_0}(1+|P|)^{-\kappa}$, after enlarging the constant on the bounded core. The upper barrier satisfies the same estimate because $-S(\iota P) = S(P)$ and $|\iota P| = |P|$. The bounds on v now give (1.2), with $\kappa = 5/2$. This proves Proposition 7.

7. Nonplanarity and the interface at infinity

Proof of Theorem 1. Let v be the minimizer from Proposition 7. The saddle value asymptotics in Section 4 and (1.2) give $v(P) - q(t(P)) \to 0$ uniformly as $|P| \to \infty$. The ray limits in the theorem concern $P = r\omega$, where $|\omega| = 1$ and $\omega \notin \Sigma$. Since $t(r\omega) = rt(\omega)$, the end limits of q give $v(r\omega) \to 1$ when $\omega \in D$ and $v(r\omega) \to -1$ when $t(\omega) < 0$.

Suppose $v(P) = p(e\cdot P)$ for a unit vector $e = (e_x, e_y)$. Choose unit vectors $\omega_x \perp e_x$ and $\omega_y \perp e_y$ in their respective four-dimensional blocks. Both rays $(r\omega_x, 0)$ and $(0, r\omega_y)$ would have the constant value $p(0)$. Their limits are -1 and 1, respectively, which proves nonplanarity.

Put $w = v - S$. By Proposition 7 and the saddle equation in Section 4, v and S are bounded entire solutions, so the interior estimates of Lemma 2 on translated unit balls bound $D^2 w$ uniformly. For $r = |P| \ge 2$ and a unit vector e, take $h = r^{-\kappa/2}$, where $\kappa = 5/2$. Taylor's formula gives

$$|\partial_e w(P)| \le \frac{|w(P+he)| + |w(P)|}{h} + \frac{h}{2}\|D^2 w\|_\infty \le C_s r^{-\kappa/2}.$$

In the last inequality, $|P+he| \ge r-1 \ge r/2$, so (1.2) bounds the numerator by $C_s r^{-\kappa}$. The estimate is uniform in e. At fixed block directions $\omega_x, \omega_y \in S^3$, the inverse formulas of Section 3 give

$$v(\rho, t) = v\left(\frac{\rho - t}{\sqrt{2}}\omega_x, \frac{\rho + t}{\sqrt{2}}\omega_y\right), \qquad |t| < \rho.$$

Block invariance makes this value independent of the directions. The map inside v is affine, so the reduced function is jointly smooth in (ρ, t), and increasing t at fixed ρ moves along the unit vector $(-\omega_x, \omega_y)/\sqrt{2}$. Applying the gradient estimate to this direction and using the saddle gradient asymptotics gives

$$\partial_t v(\rho, t) = q'(t) + o(1) \qquad \text{uniformly for } |t| \le 1 \text{ as } \rho \to \infty.$$

Let $m = \min_{[-1,1]} q' > 0$. Choose $R_* > 1$ so that $\partial_t v \geq m/2$ on this band whenever $\rho > R_*$. Since q is increasing and $q(-1) < 0 < q(1)$, the uniform value convergence also gives $v < 0$ on $t \leq -1$ and $v > 0$ on $t \geq 1$ for $\rho > R_*$, after increasing R_*.

For each $\rho > R_*$ there is thus a unique zero $t = \gamma(\rho) \in (-1, 1)$, by the opposite signs at the band endpoints and strict increase between them. The implicit function theorem gives a smooth local graph at each zero, and uniqueness makes these graphs agree on overlaps. Since $S(\rho, 0) = 0$, integrating the derivative bound between 0 and $\gamma(\rho)$ and using (1.2) gives

$$\frac{m}{2}|\gamma(\rho)| \leq |v(\rho, 0)| \leq C_s(1 + \rho)^{-5/2}.$$

At the cone point $Z = (\rho\omega_x/\sqrt{2}, \rho\omega_y/\sqrt{2})$, $\nabla t(Z) = (-\omega_x, \omega_y)/\sqrt{2}$ by differentiating the two block norms. This unit vector is perpendicular to the level set $\Sigma = \{t = 0\}$ and points into $\{t > 0\} = D$, so it is the inward normal $\nu_\Sigma(Z)$. Thus $Z + \gamma(|Z|)\nu_\Sigma(Z)$ has coordinates $(\rho, \gamma(\rho))$ and the same block directions. The set $\{\rho \leq R_*\}$ is closed by continuity of ρ and bounded by $|P| \leq \sqrt{2}\rho$, hence compact. Outside it, the zero set is exactly this normal graph over $\{Z \in \Sigma : |Z| > R_*\}$, with the stated height bound. □

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